Contents

Chapter 1

Materials and Methods

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present work). The chapter then focuses on the experimental and numerical contributions carried out during this thesis. Section 2.2 describes the calibration of the imaging systems and the experimental procedures used to characterize the cold atomic cloud, including the loading dynamics into the Optical Dipole Trap (ODT). A central focus of this experimental work is the implementation of the optical conveyor belt, which requires the precise coupling and alignment of two counter-propagating laser fields. Section 2.4 details the injection of the 1064 nm trapping light into a polarization-maintaining single-mode fiber (referred to as the "black fiber"), while Section 2.5 characterizes the power and polarization stability of this beam using Allan-deviation and frequency-domain shot-noise analyses. Section 2.6 discusses the optical layout on the upper breadboard, where the top-down black-fiber beam and the bottom-up beam emerging from the Hollow-Core Photonic Crystal Fiber (HCPCF) are spatially overlapped to form the moving optical lattice. Section 2.7 presents a full multi-level numerical model, developed using QuTiP, to simulate the Electromagnetically Induced Transparency (EIT) cooling dynamics that are required to achieve sub-Doppler temperatures once the atoms are transported inside the hollow core.

1.1 Experimental Setup

This section describes the apparatus used to produce, cool, and trap a sample of ^{87}Rb atoms. The atomic sample is prepared via a standard Magneto-Optical Trap (MOT) followed by an optical molasses phase, providing the cold atomic cloud required to load the Optical Dipole Trap (ODT). Spatial confinement and initial cooling rely on three pairs of counter-propagating, orthogonally polarized (σ^+/σ^-), red-detuned laser beams intersecting a quadrupole magnetic field. Following the MOT stage, the magnetic gradient is switched off to perform sub-Doppler cooling in an optical molasses (Sec. 1.2).

1.1.1 Setup Description

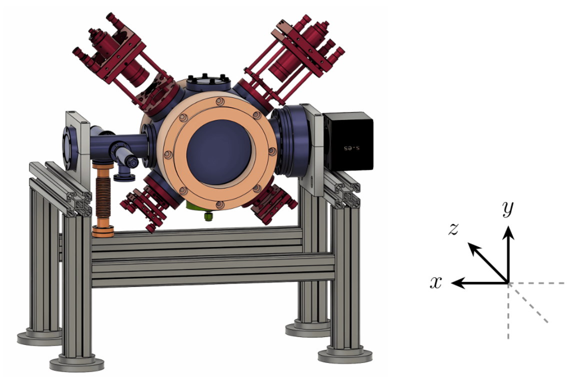


Figure 1.1: *CAD rendering of the vacuum chamber and surrounding experimental setup.*

The core of the apparatus is an Ultra-High Vacuum (UHV) chamber (Fig. 1.1) maintained at a baseline pressure of $\sim 10^{-10}$ mbar by an ion/getter combination pump (NEXTorr Z200). A heated ampoule of pure ^{87}Rb supplies the atomic vapor, raising the pressure to $\sim 10^{-8}$ mbar.

Optical access is provided by eight windows: two large CF100 viewports along the z-axis and four CF40 viewports at 45°, all AR-coated for 780 nm to optimize MOT beam transmission. Two uncoated CF40 viewports provide access for the second ODT beam and the ion pump. Due to reflections from the ion pump on the x-axis, imaging is performed through the z-axis viewports.

The MOT quadrupole magnetic field is generated by anti-Helmholtz coils (110 mm inner diameter) producing a measured axial gradient of $1.33\,\mathrm{G\,cm^{-1}A^{-1}}$ (and a corresponding radial gradient of half this value with opposite sign) [1]. Three mutually orthogonal pairs of Helmholtz shim coils cancel stray fields and are used to optimize the MOT-ODT spatial overlap, with calibration factors [2]:

$$B_z \approx 1.4 \text{ mG/mA}, \qquad B_x \approx 1.3 \text{ mG/mA}, \qquad B_y \approx 1.5 \text{ mG/mA}.$$

A Hollow-Core Photonic Crystal Fiber (HCPCF) is introduced from the bottom of the chamber via a custom mount, positioning its tip ~ 7 mm below the trap center. The optical layout is fiber-coupled using polarization-maintaining (PANDA) fibers [2]. The MOT beams are delivered via three fiber collimators and retro-reflected with quarter-wave plates to maintain the correct polarization sequence.

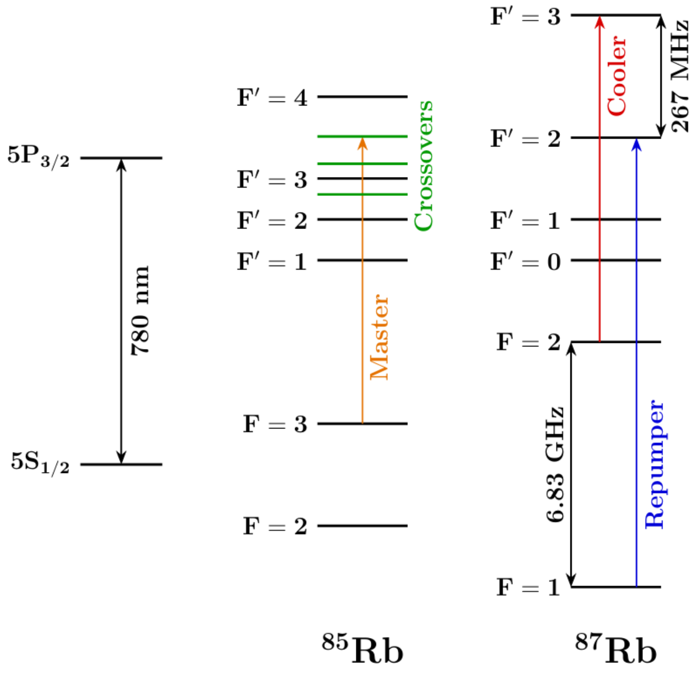


Figure 1.2: *Level scheme of the* ^{85}*Rb and* ^{87}*Rb D2 line.*

1.1.2 Laser System and Frequency Stabilization

Cooling operates on the ^{87}Rb $5^2S_{1/2} \to 5^2P_{3/2}$ (D2) transition near 780 nm (Fig. 1.2). The cooling laser drives the closed $F = 2 \to F' = 3$ transition, while a repumper laser on the $F = 1 \to F' = 2$ transition recovers atoms that off-resonantly decay into the $F = 1$ dark state. Both lasers are offset-locked to a common master laser [2].

1.1.2.1 Master laser

The master reference is a 780 nm DFB laser diode stabilized via frequency-modulated saturation spectroscopy in a heated ^{85}Rb vapor cell. It is locked to the ^{85}Rb $(F = 3 \to F' = 3) \times (F = 3 \to F' = 4)$ crossover resonance, providing a stable absolute frequency located between the ^{87}Rb cooling and repumping transitions [1].

1.1.2.2 Cooler and repumper lasers

The cooling and repumping light is generated by two 1560 nm DFB diodes. Their outputs are fiber-combined, amplified to ~ 500 mW by an Erbium-Doped Fiber Amplifier (EDFA), and frequency-doubled to 780 nm via a Second Harmonic Generation (SHG) crystal (Fig. 1.3).
Most of this light (90%) is distributed to the MOT optics via a fast-switching AOM. The remaining 10% is mixed with the master laser on a fast (~ 10 GHz) photodiode to generate beat-note signals for frequency offset locking.

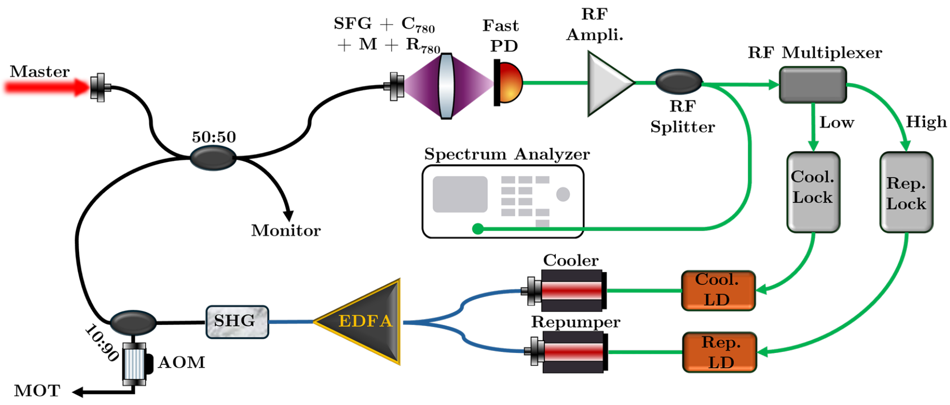


Figure 1.3: *Beat-note locking setup scheme [1].*

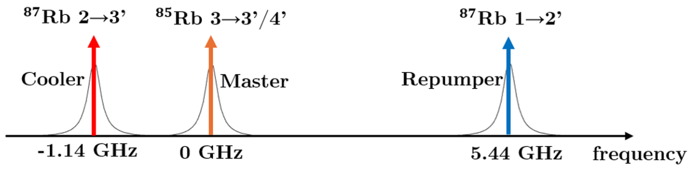


Figure 1.4: *Example of beat-note signals on a spectrum analyzer, used for frequency offset locking of the cooler and repumper lasers relative to the master laser [1].*

The master-cooler and master-repumper beat notes are isolated using RF filters and used as reference signals to stabilize the frequency offsets of the cooler and repumper lasers with respect to the master laser [1] (Figure 1.4).

1.2 Experimental Procedures and Data Analysis

1.2.1 Cold-Atom Preparation: MOT and Optical Molasses

Each experimental cycle begins with loading atoms into the magneto-optical trap (MOT), using the operating conditions previously identified as optimal for this apparatus. The MOT is run with an axial magnetic-field gradient of $B' = 9.32\,\mathrm{G/cm}$, corresponding to a current of $8\,\mathrm{A}$ in the quadrupole coils, a cooling-light detuning of $\Delta = -13\,\mathrm{MHz}$, and all compensation-coil currents set to zero, the loading stage lasts $5\,\mathrm{s}$.
At the end of the MOT phase, the quadrupole magnetic field is switched off and the cloud enters the optical molasses stage. In the absence of the trapping field, the atoms are no longer confined and the cloud starts to expand. At the same time, the atomic temperature decreases further, while the cloud falls under gravity and remains subject to a velocity-dependent damping force generated by the cooling light.
The temperature achieved during the molasses phase scales as $I/|\Delta|$, where I is the laser intensity and Δ is the detuning(Sec. 1.2.3). However, this relation remains valid only as long as $|\Delta|$ is small enough to avoid a significant reduction in the optical scattering rate, $\Gamma_p \propto 1/\Delta^2$. Furthermore, the detuning must not be so large as to induce a significant excitation probability for the neighboring $F = 2 \rightarrow F' = 2$ transition.
A second requirement for efficient molasses cooling is the cancellation of residual magnetic fields. For this reason, $5\,\mathrm{ms}$ after the MOT field is switched off, the compensation-coil currents are adjusted in order to compensate for the Earth's magnetic field and for other stray fields present in the laboratory. The cooling-light detuning is then changed from the MOT value to the molasses value through a 20-step ramp with total duration $5\,\mathrm{ms}$. This ramp starts $50\,\mu\mathrm{s}$ after the compensation currents are set to their molasses values.

1.2.2 Atom Imaging

The atomic cloud is observed inside the vacuum chamber by means of two monochrome CMOS cameras:

- a Teledyne FLIR BFS-U3-04S2M-CS camera, based on the Sony IMX287 sensor, with a resolution of 720×540 pixels and a pixel size of $6.9\,\mu\mathrm{m} \times 6.9\,\mu\mathrm{m}$;

- an Allied Vision Manta G-419 NIR camera, based on the CMOSIS/ams CMV4000 sensor, with a resolution of 2048×2048 pixels and a pixel size of $5.5\,\mu\mathrm{m} \times 5.5\,\mu\mathrm{m}$.

The FLIR camera is used mainly during the optimization of the molasses stage. The Manta camera, owing to its higher sensitivity at 780 nm and its smaller pixel pitch, is used for ODT loading measurements, where resolving the small trapped cloud is more demanding.
Both cameras are equipped with a Thorlabs MVL50M23 adjustable objective, with focal length 50 mm, minimum working distance 200 mm, and minimum magnification of about 0.33. The imaging system is focused so that the HCPCF inside the chamber, and therefore also the atomic cloud located at nearly the same distance, is sharply imaged on the sensor. In both cases the acquisition is triggered externally by the FPGA control system.

1.2.3 FLIR Camera Calibration

A calibration of the imaging system is required in order to convert distances measured on the sensor into physical lengths in the object plane. This is needed for the temperature measurements, where the cloud width extracted from the images must be expressed in absolute units. Starting from the physical pixel pitch of the FLIR sensor, 6.9 μm, the optical magnification M and the corresponding object-plane pixel size C were estimated through three independent methods.

1. **Direct imaging of a known microstructure**

   The first method uses the hollow-core photonic crystal fiber (HCPCF) as a reference object, exploiting its known physical diameter of 316 μm. The fiber is imaged directly to determine its apparent diameter on the sensor, which is then compared with its actual physical diameter. To extract the apparent diameter, the image intensity is summed along the columns, and the resulting one-dimensional profile is fitted with a rectangular function. The fitted image corresponds to an apparent diameter of 8 ± 2 pixels. The resulting object-plane calibration is therefore

   $$C = \frac{316\,\mu\text{m}}{(8 \pm 2)\,\text{px}} = 39.5 \pm 9.9\,\mu\text{m/px}, \tag{1.1}$$

   which gives a magnification

   $$M = \frac{6.9}{39.5 \pm 9.9} = 0.17 \pm 0.04. \tag{1.2}$$

2. **Optical calibration by reflected ruler imaging**

   The second method relies on imaging a macroscopic reference target placed at the same optical distance as the atomic cloud. A ruler is observed through a mirror after the camera has already been focused on the atoms. The ruler is translated until its reflected image becomes sharp, ensuring that the target lies on the same focal plane as the cloud.

In order to reduce the uncertainty and average over possible local imperfections, multiple measurements are performed. Specifically, five independent measurements are taken by repositioning the ruler at focus each time, finding that a physical interval of 2 cm corresponds to 471 ± 2 px. This procedure yields an effective pixel size

$$C = 43.4 \pm 0.3\,\mu\text{m/px}, \tag{1.3}$$

corresponding to a magnification

$$M = \frac{6.9}{43.4 \pm 0.3} = 0.159 \pm 0.001. \tag{1.4}$$

3. **Calibration from MOT displacement in a bias magnetic field**

As a further cross-check, the imaging scale can be estimated from the displacement of the MOT induced by a controlled bias magnetic field. When a uniform field is applied along the vertical direction by means of the y-axis compensation coils, the zero of the quadrupole field is shifted, and the cloud follows the new equilibrium position.

Experimentally, the cloud displacement varies linearly with the current applied to the y-axis compensation coils. Furthermore, the beam trajectory on the sensor shows a linear correlation between the horizontal and vertical displacements. Converting the fitted values from the plot into px/A, the measured slopes are

$$x_{\text{shift}} = 20.3 \pm 0.5\,\text{px/A}, \tag{1.5}$$

$$y_{\text{shift}} = 58.6 \pm 0.6\,\text{px/A}. \tag{1.6}$$

To convert this result into a physical displacement, one needs the bias field generated per unit compensation current and the restoring gradient of the quadrupole field. Modeling the compensation coils as a cubic structure of side $L = 645$ mm with $N = 100$ windings gives

$$b = 1.43\,\text{G/A}. \tag{1.7}$$

The quadrupole gradient is estimated from the coil geometry. For a current of 8.4 A, 128 windings, effective radius $R_{\text{eff}} = 62$ mm, and effective separation $a_{\text{eff}} = 162$ mm, the vertical gradient is

$$G_{y,\text{tot}} = 0.571\,\text{G/mm}. \tag{1.8}$$

The equilibrium displacement per unit current is given by the ratio between the applied bias field and the magnetic-field gradient. Dividing this physical shift by the measured vertical displacement in pixels gives the calibration factor:

$$C = \frac{b}{G_{y,\text{tot}}\, y_{\text{shift}}} = \frac{1.43\,\text{G/A}}{0.571\,\text{G/mm} \times (58.6 \pm 0.6)\,\text{px/A}} = 42.7 \pm 0.4\,\mu\text{m/px}. \tag{1.9}$$

If, instead, the empirically measured values previously obtained for this setup are used [2] namely $b = 1.5\,\text{G/A}$ and $G = 5.6\,\text{G/cm}$ at $8.4\,\text{A}$, one obtains

$$C = \frac{1.5\,\text{G/A}}{5.6\,\text{G/cm} \times (58.6 \pm 0.6)\,\text{px/A}} = 45.7 \pm 0.5\,\mu\text{m/px}. \tag{1.10}$$

Both estimates are consistent with the result obtained from the ruler calibration. In the following analysis, the value provided by the ruler method is adopted.

A summary of the three calibration methods is shown in Fig. 1.5.

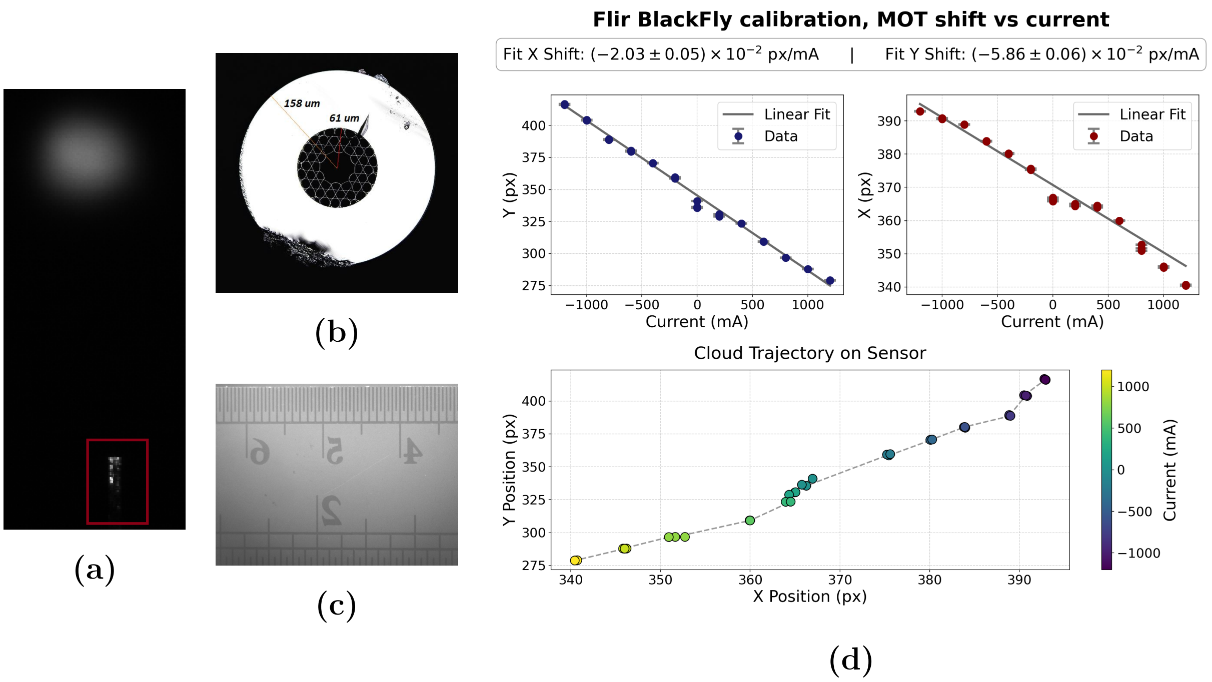


Figure 1.5: *Summary of the spatial calibration methods used for the imaging system.* ***(a)*** *Side-view image of the HCPCF used as the reference object in Method 1.* ***(b)*** *Cross-sectional view of the same fiber, with the nominal diameter of* $316\,\mu\text{m}$. ***(c)*** *Ruler used for the reflected-image calibration of Method 2.* ***(d)*** *MOT position as a function of the current applied to the y-axis compensation coils in Method 3; the linear fit yields a slope of about* $58\,\text{px/A}$.

1.2.4 Temperature Measurements

Once the imaging calibration is known, the cloud size extracted from the images can be expressed in physical units and used to determine the temperature from ballistic expan-

sion. These measurements are used to optimize the molasses parameters by identifying the conditions that minimize the final temperature.
The method is based on the free expansion of the cloud after release. For a non-interacting gas, the position of an atom along one direction evolves as

$$x(t) = x_0 + v_x t. \tag{1.11}$$

Assuming that the initial position and velocity are statistically independent, the variance at time t is

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{1.12}$$

where $\sigma_{x,0}$ is the initial spatial width. In the center-of-mass frame, the equipartition theorem gives

$$\sigma_{v_x}^2 = \langle v_x^2 \rangle = \frac{k_B T_x}{m}. \tag{1.13}$$

The fitting law then becomes

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2, \tag{1.14}$$

with m the mass of ^{87}Rb, $m \approx 1.443 \times 10^{-25}$ kg.
The experimental procedure is as follows:

1. A MOT is loaded and followed by the molasses stage. The molasses parameters and duration are the quantities under optimization.

2. The cooling light is switched off, allowing the cloud to expand freely and fall under gravity. At each selected time of flight, the FLIR camera acquires a fluorescence image of the cloud.

3. A background frame is then acquired after a sufficiently long delay, so that the cloud has completely left the observation region.

4. For each time of flight, the corrected image is obtained as

$$I(t) = I_{\text{raw}}(t) - I_{\text{bg}}(t). \tag{1.15}$$

5. Each corrected image is fitted with a two-dimensional Gaussian, yielding the widths σ_x and σ_y.

6. Equation 1.14 is fitted independently along the two axes to extract T_x and T_y. The temperature quoted in the following is the average of the two values.

The imaging process is destructive because the light flash heats the sample, resulting in its loss. Therefore, each time-of-flight measurement cannot be done sequentially, but requires a new experimental run from start to finish.

1.2.5 ODT Loading Measurements

The loading performance of the optical dipole trap is characterized using the Manta camera. Two quantities are extracted from these measurements: the loading efficiency η and the integrated fluorescence signal associated with the trapped cloud, which is proportional to the number of captured atoms.
Because only a small fraction of the molasses atoms, of the order of 10^{-3}, is transferred into the ODT, the trapped-atom signal is much weaker than the fluorescence produced by the untrapped cloud. For this reason, the analysis is based on differential imaging.

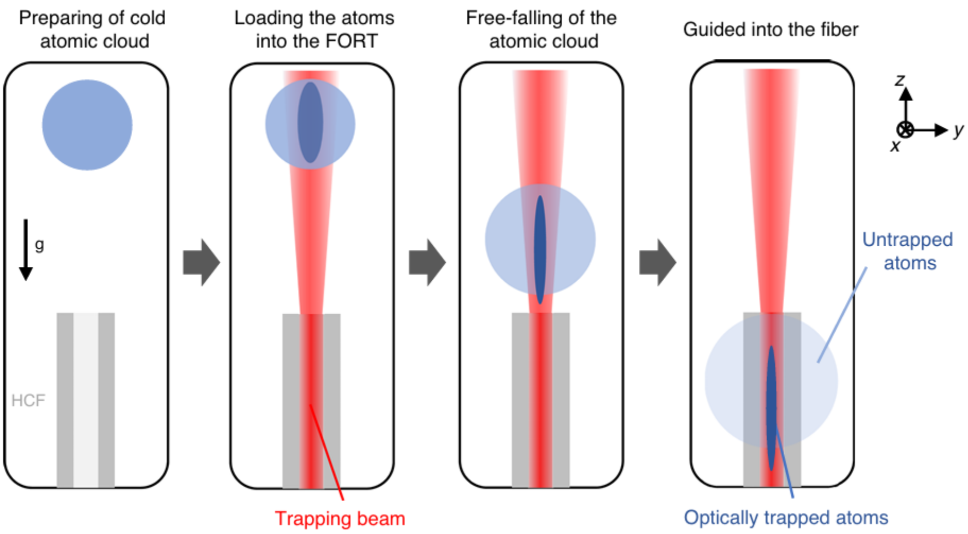


Figure 1.6: *Schematic representation of the experimental sequence for loading and guiding atoms. The process consists of four stages: (1) preparation of the cold atomic cloud, (2) loading the atoms into the far-off-resonance trap (FORT) using a trapping beam, (3) free-falling of the atomic cloud under gravity, and (4) guiding of the optically trapped atoms into the hollow-core fiber (HCF), while the untrapped atoms fall and disperse outside the trap. Credits [3].*

The measurement sequence is the following (illustrated in Fig. 1.6):

1. Two otherwise identical experimental cycles are performed in succession, one with the ODT on from the MOT loading stage onward and one with the ODT off.

2. After a time of flight of 30 ms, long enough for the untrapped atoms to expand away from the ODT region, fluorescence images I_{on} and I_{off} are acquired with the Manta camera.

3. Each on/off pair is repeated five times.

4. The images are cropped to a pixel region of interest centered on the cloud. This excludes both the visible HCPCF structure and the peripheral parts of the image containing only background.

5. For each condition, the cropped images are averaged:

$$\bar{I}_{\mathrm{on}} = \langle I_{\mathrm{on}} \rangle, \qquad \bar{I}_{\mathrm{off}} = \langle I_{\mathrm{off}} \rangle. \tag{1.16}$$

6. The averaged images are integrated along one direction to obtain one-dimensional profiles with improved signal-to-noise ratio.

7. The total number of atoms in the molasses drifts slowly over time because of variations in vacuum pressure and thermal changes in the optical components. As a consequence, the on and off profiles cannot be subtracted directly: even a small mismatch in the broad molasses background would produce a residual comparable to the ODT signal itself.

   To correct for this effect, the off profile is rescaled by a factor k chosen so that the difference between the two profiles vanishes outside the ODT region. An exclusion window centered at the ODT position, $x_{\mathrm{ODT}} \pm R_{\mathrm{window}}$, is defined. Outside this interval, a linear baseline $L_i(x)$ is estimated for each profile and the residual integrated signal is computed as

$$A_i^{\mathrm{out}} = \int_{\mathrm{outside}} \left[\bar{I}_i(x) - L_i(x) \right] dx, \qquad i \in \{\mathrm{on}, \mathrm{off}\}. \tag{1.17}$$

   The normalization factor is then

$$k = \frac{A_{\mathrm{on}}^{\mathrm{out}}}{A_{\mathrm{off}}^{\mathrm{out}}}. \tag{1.18}$$

8. The differential profile attributed to the trapped atoms is therefore

$$\Delta \bar{I}(x) = \bar{I}_{\mathrm{on}}(x) - k\, \bar{I}_{\mathrm{off}}(x). \tag{1.19}$$

9. A Gaussian plus linear background is fitted both to the off profile and to the differential profile. If $A_{\mathrm{off}}^{\mathrm{G}}$ and $A_{\mathrm{ODT}}^{\mathrm{G}}$ denote the Gaussian areas extracted from the two fits, the loading efficiency is defined as

$$\eta = \frac{A_{\mathrm{ODT}}^{\mathrm{G}}}{k\, A_{\mathrm{off}}^{\mathrm{G}}}. \tag{1.20}$$

1.2.6 Measurement of the Number of Atoms

The loading efficiency can be obtained from relative fluorescence signals and therefore does not require an absolute calibration of the camera response. Estimating the absolute number of trapped atoms requires converting the recorded counts into a known number of detected photons.

To this end, the camera is calibrated against a beam of known optical power coming from the MOT light. The beam is directed onto the sensor after passing through a series of attenuators and through the vacuum-chamber viewports. The total transmission of the optical path is written as the product of the individual transmission factors,

$$T_{\mathrm{tot}} = T_1\, T_{\mathrm{vac}}\, T_2\, T_3 \approx 0.01272. \tag{1.21}$$

During the calibration, the AOM amplitude is set to 50, corresponding to an input power of 22 μW after background subtraction. The optical power reaching the camera is therefore

$$P_{\mathrm{cam}} = 22\,\mu\mathrm{W} \times T_{\mathrm{tot}} \approx 279.8\,\mathrm{nW}. \tag{1.22}$$

The camera response is determined by varying the exposure time t_{exp} while keeping P_{cam} fixed. For each image, the beam profile is fitted in two dimensions and the integrated counts are extracted. Denoting by C the total number of recorded counts, the response is assumed to be linear:

$$C = m\, t_{\mathrm{exp}} + q, \tag{1.23}$$

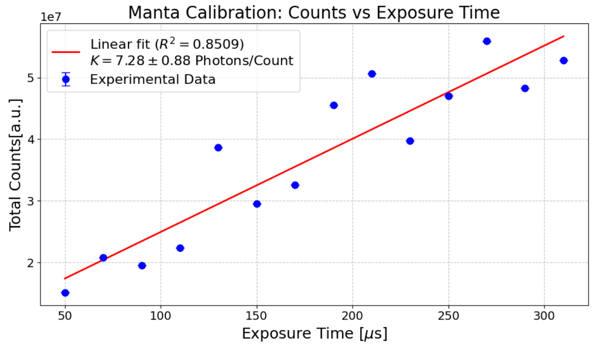


Figure 1.7: *Linear fit of the total camera counts as a function of the exposure time* t_{exp}. *The slope* m *extracted from the fit is used to calculate the photon conversion factor* K.

where m is the count rate. Since the incident photon flux is given by $P_{\mathrm{cam}}/(hc/\lambda)$, the conversion factor from counts to detected photons is

$$K = \frac{P_{\mathrm{cam}}\lambda}{mhc}, \tag{1.24}$$

where h is Planck's constant, c the speed of light, and λ the laser wavelength. The resulting linear fit is shown in Fig. 1.7, from which the conversion factor K is extracted.

For an ODT fluorescence image, the integrated residual signal C_{tot} can then be converted into the number of detected photons:

$$N_{\mathrm{det}} = K\, C_{\mathrm{tot}}. \tag{1.25}$$

In order to infer the number of trapped atoms, the fluorescence contribution of a single atom during the imaging pulse must be estimated. The scattering rate is described by the standard two-level expression

$$R_{\mathrm{sc}} = \frac{\Gamma}{2} \frac{I/I_{\mathrm{sat}}}{1 + I/I_{\mathrm{sat}} + 4(\Delta/\Gamma)^2}, \tag{1.26}$$

where Γ is the natural linewidth, I is the total imaging intensity, I_{sat} is the saturation intensity, and Δ is the laser detuning. Considering a single MOT beam with a power of $P = 14\,\mathrm{mW}$ and a waist of $W = 12.5\,\mathrm{mm}$, the peak intensity is $I_0 = 2P/(\pi W^2) \approx 5.70\,\mathrm{mW/cm^2}$. Using the ideal two-level saturation intensity for the ^{87}Rb cooling transition ($I_{\mathrm{sat}} = 1.67\,\mathrm{mW/cm^2}$), the saturation ratio is evaluated as $s_0 = I_0/I_{\mathrm{sat}} \approx 3.4$.
For an imaging pulse of duration t_{img}, each atom scatters on average $R_{\mathrm{sc}} t_{\mathrm{img}}$ photons. Only a fraction of this fluorescence reaches the sensor. Approximating the collection efficiency by the fractional solid angle subtended by the imaging optics,

$$\eta_{\mathrm{coll}} = \frac{r^2}{4d^2}, \tag{1.27}$$

with $r = 20\,\mathrm{mm}$ the effective lens radius and $d = 430\,\mathrm{mm}$ the distance between the cloud and the collection optics, obtaining a geometrical efficiency of:

$$\eta_{\mathrm{coll}} = \frac{20^2}{4(430)^2} \approx 5.41 \times 10^{-4}. \tag{1.28}$$

The number of detected photons per atom is

$$N_{\mathrm{ph},1} = R_{\mathrm{sc}}\, t_{\mathrm{img}}\, \eta_{\mathrm{coll}}\, T_{\mathrm{tot}}, \tag{1.29}$$

where T_{tot} accounts for the transmission of the imaging system.
The absolute number of trapped atoms is finally obtained from

$$N = \frac{N_{\mathrm{det}}}{N_{\mathrm{ph},1}} = \frac{K\, C_{\mathrm{tot}}}{R_{\mathrm{sc}}\, t_{\mathrm{img}}\, \eta_{\mathrm{coll}}\, T_{\mathrm{tot}}}. \tag{1.30}$$

1.3 Expected Number of Trapped Atoms

A simple estimate of the number of atoms loaded into the ODT can be obtained by combining the spatial density distribution of the MOT with the thermal velocity distribution of the atoms. An atom at position $\vec{r}$ is assumed to be captured if its transverse kinetic energy is smaller than the local trap depth:

$$\frac{1}{2} m \left(v_x^2 + v_y^2\right) < \left|U_{\mathrm{ODT}}(\vec{r})\right|. \tag{1.31}$$

The spatial density of the initial cloud is denoted by $n_{\mathrm{MOT}}(\vec{r})$, while the velocity distribution is taken to be Maxwell-Boltzmann,

$$f(\vec{v}) = \frac{1}{(2\pi\mu^2)^{3/2}} \exp\left(-\frac{v^2}{2\mu^2}\right), \qquad \mu^2 = \frac{k_B T}{m}. \tag{1.32}$$

The number of captured atoms is obtained by integrating over phase space, restricting the result to atoms moving toward the trap region, namely those with $v_z < 0$. Under this assumption,

$$N_{\mathrm{ODT}} = \int d^3r \, n_{\mathrm{MOT}}(\vec{r}) \, \frac{1}{2} \int_{v_x^2+v_y^2<\frac{2|U|}{m}} dv_x \, dv_y \, \frac{1}{2\pi\mu^2} \exp\left(-\frac{v_x^2+v_y^2}{2\mu^2}\right). \tag{1.33}$$

The velocity integral is more conveniently evaluated in polar coordinates in the transverse velocity plane. Using

$$dv_x \, dv_y = \pi \, d(v_r^2), \tag{1.34}$$

and introducing the variable

$$x = \frac{v_r^2}{2\mu^2}, \tag{1.35}$$

one obtains

$$\int_0^{\frac{2|U|}{m}} \pi \, d(v_r^2) \, \frac{1}{2\pi\mu^2} \exp\left(-\frac{v_r^2}{2\mu^2}\right) = \int_0^{\frac{|U|}{m\mu^2}} e^{-x} \, dx = 1 - \exp\left(-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_B T}\right). \tag{1.36}$$

The expression for the number of trapped atoms therefore becomes

$$N_{\mathrm{ODT}} = \int d^3r \, n_{\mathrm{MOT}}(\vec{r}) \, \frac{1}{2} \left[1 - \exp\left(-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_B T}\right)\right]. \tag{1.37}$$

In the shallow-trap limit, $|U_{\mathrm{ODT}}| \ll k_B T$, the exponential can be expanded to first order one then finds:

$$N_{\mathrm{ODT}} \simeq \int d^3r \, n_{\mathrm{MOT}}(\vec{r}) \, \frac{|U_{\mathrm{ODT}}(\vec{r})|}{2k_B T}. \tag{1.38}$$

For a Gaussian dipole trap,

$$U_{\mathrm{ODT}}(\vec{r}) = -U_0 \, \frac{W_0^2}{W^2(z)} \exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right), \tag{1.39}$$

so that

$$N_{\mathrm{ODT}} \simeq \frac{U_0}{2k_B T} \int d^3r \, n_{\mathrm{MOT}}(\vec{r}) \, \frac{W_0^2}{W^2(z)} \exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right). \tag{1.40}$$

If the ODT waist is much smaller than the transverse size of the MOT (if $W(z) \ll R_x, R_y$), the transverse integrals can be extended to infinity:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right)dx\,dy=\frac{\pi W^2(z)}{2}. \tag{1.41}$$

Substituting this result into the previous expression leads to the cancellation of the $W^2(z)$ term:

$$N_{\mathrm{ODT}}=\frac{U_0}{2k_BT}\frac{\pi W_0^2}{2}\int_{-R_z}^{R_z}n_{\mathrm{MOT}}(0,0,z)\,dz. \tag{1.42}$$

To obtain a compact analytical estimate, the MOT is modeled as a uniform ellipsoid with radii R_x, R_y, and R_z, and constant density

$$n_{\mathrm{MOT}}=\frac{3N_{\mathrm{MOT}}}{4\pi R_xR_yR_z}. \tag{1.43}$$

Under this approximation,

$$N_{\mathrm{ODT}}=\frac{U_0\pi W_0^2}{4k_BT}\left(\frac{3N_{\mathrm{MOT}}}{4\pi R_xR_yR_z}\cdot 2R_z\right)=\frac{3}{8}\frac{U_0W_0^2}{k_BTR_xR_y}N_{\mathrm{MOT}}. \tag{1.44}$$

1.4 Injection into the black fiber

The second optical field required for the conveyor-belt configuration is delivered by a conventional polarization-maintaining single-mode fiber, referred to throughout this work as the ”black fiber”. It is generated from the same 1064 nm laser but counterpropagates with respect to the field guided through the Hollow-Core Photonic Crystal Fiber (HCPCF). Their interference generates the moving optical lattice described in Sec. 1.3. Efficient coupling, a well-defined spatial mode, and stable polarization are therefore required to obtain a deep and reproducible conveyor-belt potential.

The black-fiber branch was constructed as an extension of the Optical Dipole Trap (ODT) system described already existing. The injection strategy was first validated on a dedicated test bench and subsequently transferred to the definitive optical branch.

The black fiber is operated at 1064 nm and has a nominal mode-field radius of approximately 5.2 μm. Its input tip includes a glass endcap approximately 300 μm long, which improves power handling. Specifically, the fiber is a polarization-maintaining, single-mode 3 m cable manufactured by Schäfter+Kirchhoff (model PMC-780-1-18-xxx / PMC-780-5.3-NA012-1-APC-xxx-P, housed in a 900 μm buffer). It features low-stress FC/APC (8°) connectors on both ends with the polarization slow axis aligned to the connector index key, and a specified operating wavelength range from the 770 nm cut-off up to a maximum of 1100 nm.

1.4.1 Gaussian-beam characterization procedure

Throughout this work, the size and divergence of the optical beam are characterized using a common beam-profiling procedure. The same method is applied before and after the fiber whenever the injection geometry, input beam parameters, or output wavefront have to be determined. The beam under test is first aligned horizontally at a fixed height above the optical table. This allows its transverse intensity distribution to be sampled at a sequence of longitudinal positions without introducing additional optics between measurements. A camera mounted on a translation stage is then moved along the propagation direction, and an image of the beam cross-section is recorded at each position.

Each recorded image is cropped to isolate the beam spot from background light and stray reflections. The resulting intensity distribution is fitted with a two-dimensional Gaussian function, from which the beam radii along the two transverse axes are extracted. Repeating this analysis at several positions provides the beam radius as a function of the longitudinal coordinate z. For each transverse direction, the measured beam radius is fitted to the standard Gaussian-beam propagation law

$$w(z) = w_0\sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2}, \tag{1.45}$$

where w_0 is the waist radius, z_0 is the waist position, and z_R is the associated Rayleigh range, defined as

$$z_R = \frac{\pi w_0^2}{\lambda}, \tag{1.46}$$

with λ being the wavelength of the beam. This fit provides the size and position of the beam waist and properties at any relevant plane to be determined. The reconstructed beam parameters could then be compared with the target fiber-mode parameters or used to select and position the injection optics.

1.4.2 Injection setup

The injection setup for the black fiber comprises the pre-existing ODT injection setup, shared with the HCPCF branch, and a branch built specifically for the black fiber downstream of PBS2. To the first setup a half-wave plate was added for the polarization control required to implement the conveyor belt.
The pre-existing setup, shown in Figure 1.8, consists of:

- A half-wave plate (HWP 1) and an optical isolator protecting the laser from back-reflections;
- HWP 2 and PBS 1, regulating the optical power delivered to the rest of the setup;
- HWP 3 and PBS 2, splitting the beam between the HCPCF branch and the black-fiber branch;
- on the HCPCF branch, fed by the first output of PBS2: an acousto-optic modulator (AOM) operating at 110 MHz, used as a fast switch, with the first diffraction order used downstream and the zeroth order blocked by a beam dump;
- a $\sim 3\times$ beam expander, collimating and enlarging the beam;
- a mirror (Mirror 1) and a Dichroic Mirror, both on tilt stages, used for the fiber-injection alignment, with the dichroic mirror also combining the 1064 nm beam with a 780 nm beam used for atom characterization inside the HCPCF core;
- a focusing lens with $f = 50$ mm (Lens f50), on a horizontal micrometric translation stage;
- HWP 4 for the polarization adjustment before the HCPCF, mounted on a rotation stage;

- the HCPCF (Hollow core fiber) on an x-y translation stage terminated with an FC/APC connector.

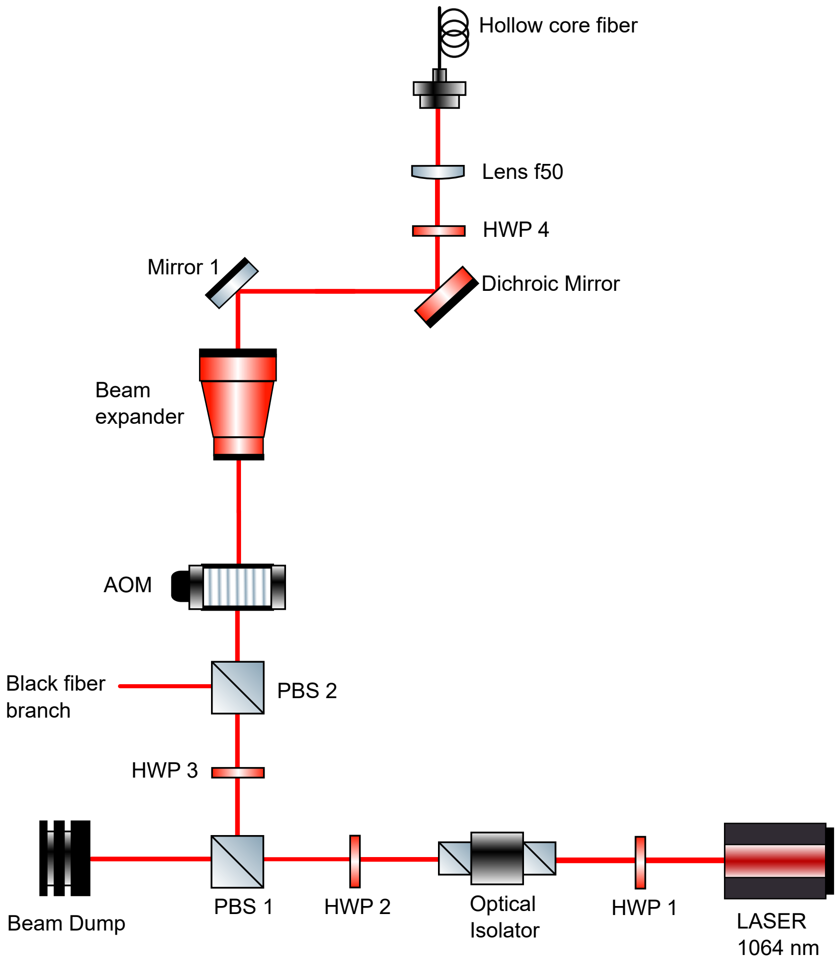


Figure 1.8: *Optical scheme of the pre-existing ODT injection setup, from HWP1 to the HCPCF, with the second output of PBS2 leading to the black-fiber branch described below.*

Downstream of PBS2, the branch built for the black fiber, shown in Figure 1.9, consists of:

- an AOM, providing fast control of the beam intensity and frequency operating at 110 MHz (its frequency is shifted to allow for the transport of the atoms in the moving lattice, as described in Section 1.3);
- a mirror (Mirror 2), directing the beam into the telescope;
- a telescope formed by an $f = -30$ mm lens (Lens f-30) and an $f = 150$ mm lens (Lens f150), re-collimating the beam to a radius of approximately 1.6 mm, the target value derived in Section 1.4.3;
- two mirrors (Mirror 3 and Mirror 4), directing the beam toward the fiber-injection assembly;

- a half-wave plate (HWP 4), setting the incident linear polarization along the slow axis of the fiber, as described in Section 1.5);
- an $f = 25\,\mathrm{mm}$ achromatic doublet (Lens f25) (Thorlabs AC127-025-C), mounted on a micrometric translation stage (z-stage) along the propagation axis, focusing the beam into the fiber. The axial position of the lens is used to compensate for the displacement of the fiber-mode plane caused by the fiber endcap, discussed in Section 1.4.3;
- the black fiber mounted on the branch through its output connector.

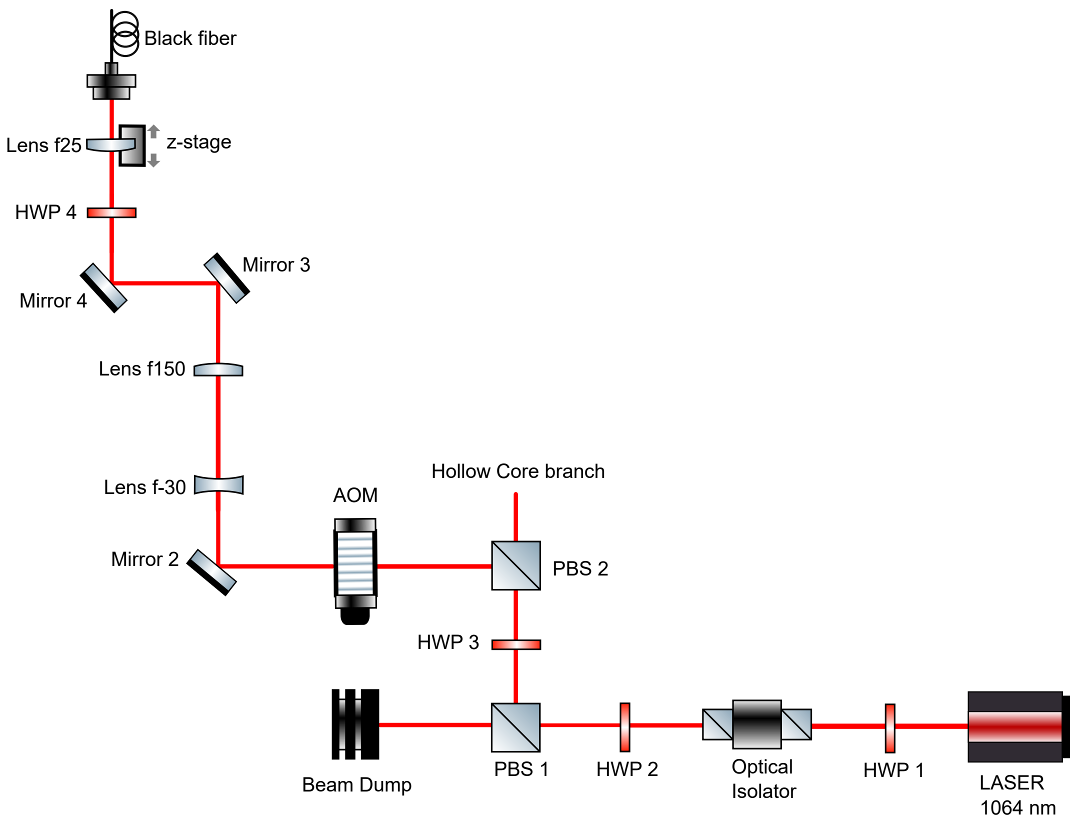


Figure 1.9: *Optical scheme of the black-fiber branch downstream of PBS2.*

Figure 1.10 shows a photograph of the complete injection setup.
The size and divergence of the beams involved in the black-fiber injection were characterized with the same beam-profiling approach used in Section 1.4.1.
For each transverse direction, the measured radius was fitted to Eq. 1.45, this fit gave the size and position of the beam waist, which could then be compared with the target fiber-mode parameters or used to select and position the injection optics. The procedure was repeated both upstream and downstream of the fiber, after each modification or addition of the injection optics, to obtain the beam parameters used in the mode-matching calculations of Section 1.4.3.

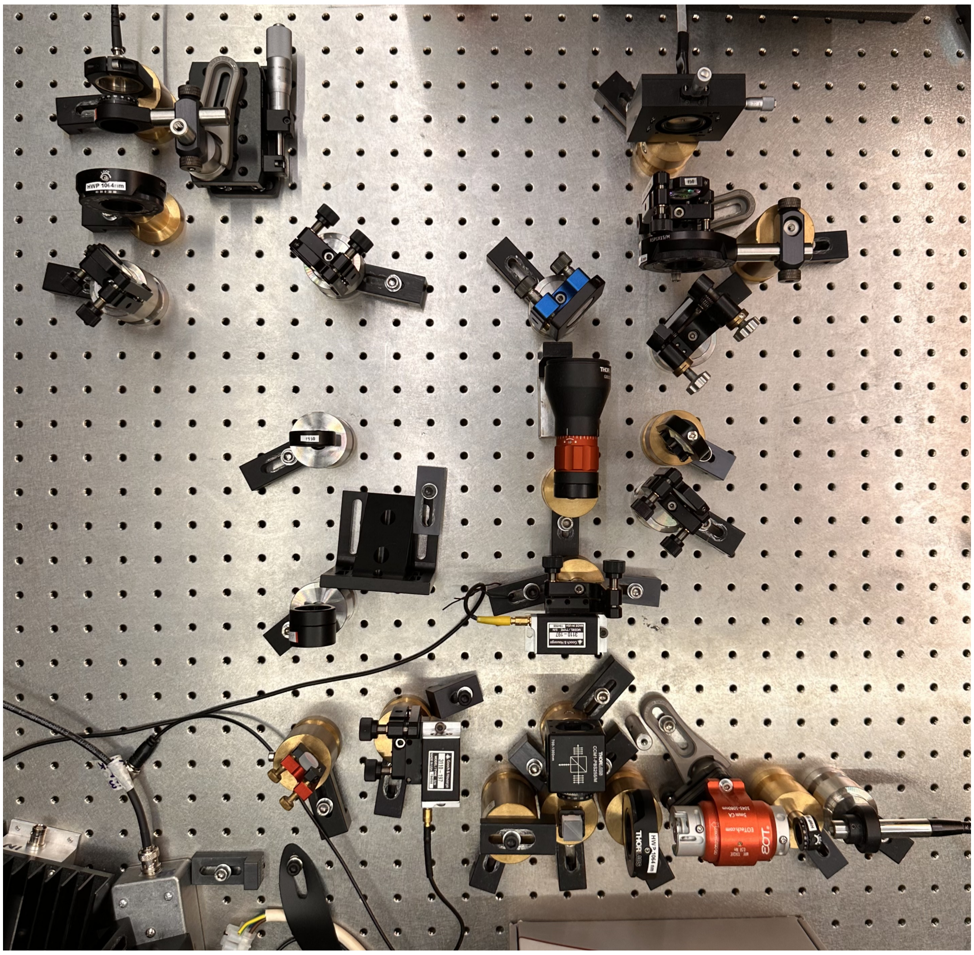


Figure 1.10: *Photograph of the complete injection branch. The black-fiber branch is on the left, while the pre-existing ODT injection setup is on the right.*

1.4.3 Mode matching and the fiber endcap

Coupling a Gaussian beam into a single-mode fiber requires matching the focused field to the fundamental fiber mode in waist size, waist position and transverse alignment. The standard approach uses a collimated input beam and a fixed collimating lens that focuses it directly onto the fiber facet, and this was the strategy first used for the black fiber. The mode-matching parameters used in the definitive branch (Section 1.4.2) were derived and validated on a dedicated test bench (Fig. 1.11, only the input section upstream of the fiber in the picture is relevant for the mode-matching and injection procedure. The optical layout at the fiber output is dedicated exclusively to the characterization of the polarization stability, which is discussed in detail in Sec. 1.5). The test bench consisted of:

- HWP 1 and the optical isolator, as in Section 1.4.2;
- HWP 2 and PBS 1 (with a beam dump on the rejected port), regulating the optical power delivered to the rest of the setup;

- HWP 3 and PBS 2, splitting the beam between the HCPCF and black-fiber branches;
- Mirror 1, directing the beam reflected by PBS2 towards the beam expander (Beam expander);
- an adjustable beam expander (ZBE3C2X-8X Achromatic Zoom Beam Expander, AR Coated: 1050 nm-1650 nm), setting the diameter and residual divergence of the beam sent to the black fiber;
- HWP 4 and PBS 3 (with a beam dump on the rejected port), for additional power control and polarization cleaning;
- Mirror 2 and Mirror 3, routing the beam towards the fiber injection stage;
- HWP 5, adjusting the polarization axis before injection into the polarization-maintaining fiber;
- a focusing element coupling the beam into the black fiber: initially a fixed $f = 18\,\mathrm{mm}$ fiber collimator (Collimator f18) (Schäfter+Kirchhoff, 60FC-4-A18-03), later replaced by the $f = 25\,\mathrm{mm}$ achromatic doublet (Lens f25) (Thorlabs AC127-025-C) on a micrometric translation stage (z-stage) also used in the definitive branch;

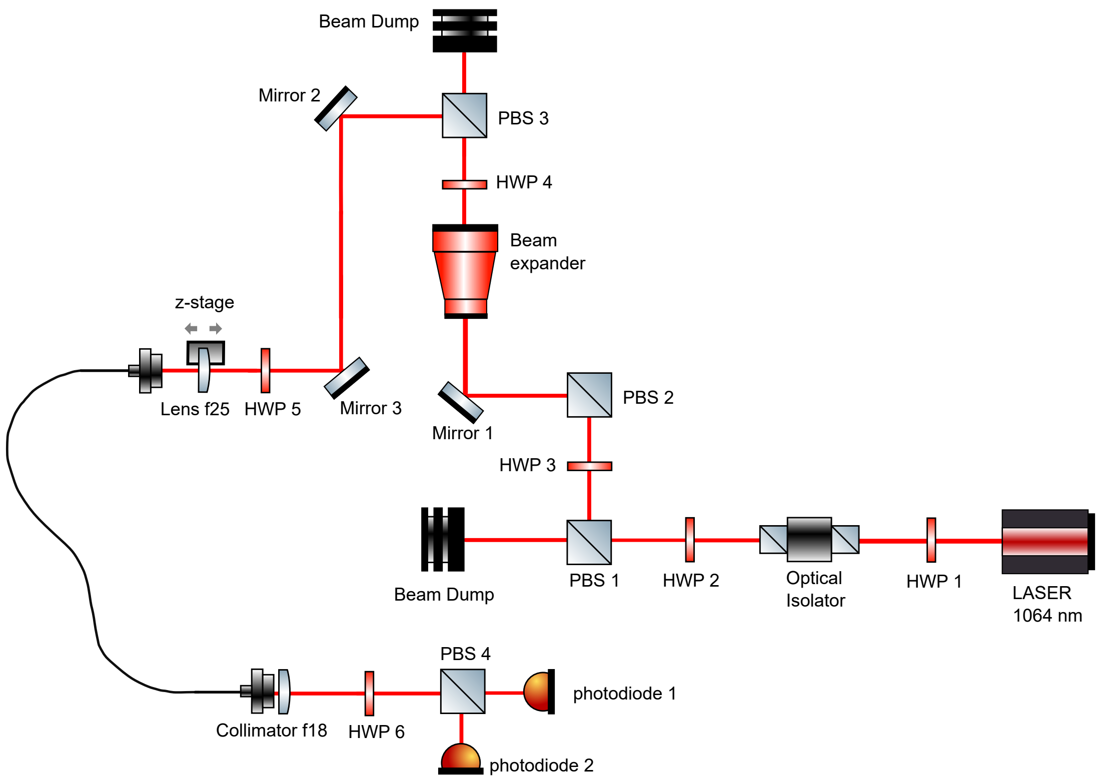
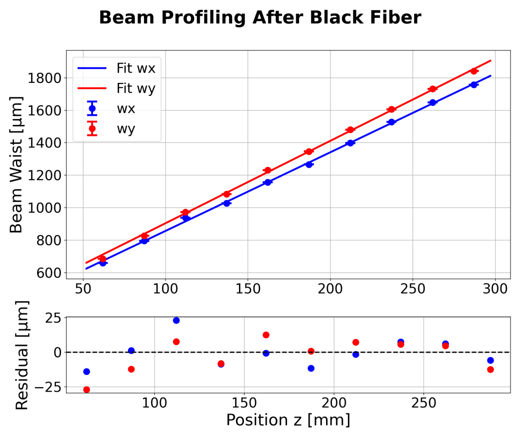
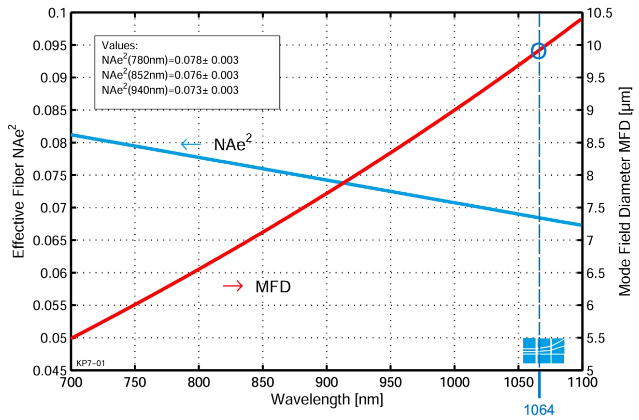


Figure 1.11: *Setup of the black-fiber injection test bench, used to validate the mode-matching procedure and calibrate the axial position of the $f = 25\,\mathrm{mm}$ lens. At the fiber output the polarization stability was characterized with the setup of Section 1.5.*

Injecting a nominally collimated beam directly through the fixed $f = 18\,\mathrm{mm}$ collimator gave an injection efficiency of approximately 20%. Introducing additional divergence upstream, equivalent to inserting a weak $f = -1000\,\mathrm{mm}$ lens before the collimator, raised the efficiency above 50%. This indicates that the plane of the fiber mode does not coincide with the nominal focal plane of the collimator. The beam profile after the fiber was measured (Figure 1.12a) and its reconstructed waist compared with the mode-field diameter (MFD) specified by the manufacturer at 1064 nm (Figure 1.12b).



(a) *Beam-radius measurements after the black fiber and fit to the Gaussian-beam propagation law.*



(b) *Manufacturer-specified mode-field diameter of the black fiber at* 1064 nm, *used as the target value in the mode-matching calculation below.*

Figure 1.12: *Black fiber beam characterization.*

The beam profile analysis shown in Fig.1.12a uses the standard Gaussian beam propagation model. The beam radius evolution $w_i(z)$ along the propagation axis z for both transverse directions ($i \in \{x, y\}$) is described by:

$$w_i(z) = w_{0i}\sqrt{1 + \left(\frac{z - z_{0i}}{z_{Ri}}\right)^2}, \quad \text{with} \quad z_{Ri} = \frac{\pi w_{0i}^2}{M_i^2 \lambda}. \tag{1.47}$$

Based on the fit of the experimental data, and keeping the beam quality factors fixed to $M_x^2 = M_y^2 = 1.02$, the extracted parameters and the respective coefficients of determination (R^2) are:

$$z_{0x} = 3.97 \pm 0.43\,\mathrm{mm}, \qquad z_{0y} = 3.85 \pm 0.48\,\mathrm{mm},$$
$$w_{0x} = (5.67 \pm 0.10) \times 10^{-3}\,\mathrm{mm}, \qquad w_{0y} = (5.42 \pm 0.11) \times 10^{-3}\,\mathrm{mm},$$
$$R_x^2 = 0.9993, \qquad R_y^2 = 0.9991.$$

The measured waist agrees with the specified MFD, confirming that the transmitted light corresponds to the fundamental fiber mode.

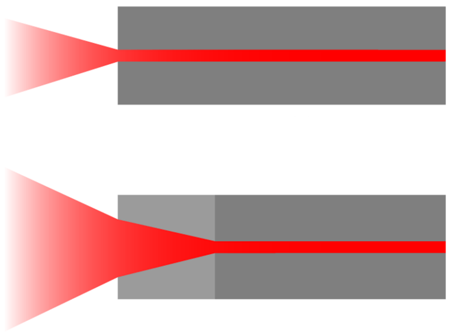


Figure 1.13: *Geometry of the endcap spliced onto the black-fiber input. Image courtesy of Thorlabs, Inc.*

The mismatch between the nominal collimator focal plane and the fiber-mode plane is attributed to the endcap spliced onto the fiber input, a glass segment approximately 300 μm long used to improve power handling (Figure 1.13). The endcap modifies the effective optical path between the external focusing lens and the fiber facet, displacing the focused waist relative to the position expected for a bare fiber tip. Standard fiber collimators are designed assuming the fiber mode lies at their nominal focal plane, so this displacement is not compensated automatically and must be corrected in the injection design. Reproducing the correction with a fixed, long-focal-length diverging lens would be impractical to position and would add aberrations to the injected beam. A focusing lens mounted on a micrometric translation stage, giving direct control of the axial position of the focused waist, was used instead.

For a collimated beam of radius w_{in} focused by a lens of focal length f, Equation (1.48) gives the resulting waist radius w_{f}. Solving for the input radius required to obtain a target w_{f},

$$w_{\text{in}} = \frac{\lambda f}{\pi w_{\text{f}}}. \tag{1.48}$$

Using $\lambda = 1.064\,\mu$m, $f = 25\,$mm, and $w_{\text{f}} = 5.2\,\mu$m, taken as half of the manufacturer-specified MFD (Figure 1.12b), gives

$$w_{\text{in}} = \frac{(1.064 \times 10^{-6}\,\text{m})(25 \times 10^{-3}\,\text{m})}{\pi(5.2 \times 10^{-6}\,\text{m})} \approx 1.6\,\text{mm}. \tag{1.49}$$

The beam expander was set to produce this radius at the injection lens, and the resulting beam was profiled before injection to verify both its size and its residual divergence.

A temporary reverse-direction setup was implemented for alignment purposes(not shown in Figure 1.11). The axial position of the $f = 25\,$mm lens was calibrated by propagating light in the reverse direction, using a MOT beam injected through a fixed $f = 18\,$mm collimator. The light was collected by the $f = 25\,$mm lens, and its position was adjusted until the resulting beam was collimated. This confirmed the effective fiber-mode plane and fixed the lens position for the definitive setup.

On the test bench, with the $f = 25\,$mm lens at the calibrated position and forward propagation, the power measured after the black fiber and the $f = 18\,$mm output collimator reached approximately 70% of the power measured before the fiber, directly after the $f = 25\,$mm lens.

With the definitive branch of Section 1.4.2, after adding the AOM and the telescope, the measured injection efficiency reached approximately 75%,confirming that the mode-matching procedure transfers to the complete optical layout.

1.5 Power and polarization stability characterization

After optimizing the injection efficiency, both the overall optical power and the polarization stability of the light transmitted through the black fiber were characterized. These quantities are relevant to the optical conveyor belt because polarization fluctuations are converted into optical-power fluctuations by the polarization-sensitive elements of the final setup, ultimately affecting the stability of the resulting optical potential. The two orthogonal linear-polarization components at the fiber output were measured simultaneously. Their time-domain stability was analyzed by computing the overlapping Allan deviation of their fractional fluctuations, while their frequency content was evaluated through the Amplitude Spectral Density (ASD) and compared with the expected fundamental shot-noise limit.

1.5.1 Detection setup and PBS characterization

Before being integrated into the analysis setup, the polarizing beam splitter (PBS) under test was carefully characterized. The test setup consisted of a half-wave plate (HWP) followed by an auxiliary PBS, of the same type as the one under test, to prepare a purely linear polarization state. By measuring the transmitted and reflected power, the Polarization Extinction Ratio (PER) was evaluated, defined as:

$$\mathrm{PER} = 10\log_{10}\left(\frac{P_{\max}}{P_{\min}}\right). \tag{1.50}$$

The component exhibited a measured PER of 31 dB in transmission and 30 dB in reflection. Because the auxiliary PBS used to prepare the input state has a finite extinction ratio itself, the incident polarization is not perfectly linear. Consequently, these measured values represent a lower limit for the actual performance of the component, confirming its suitability for high-sensitivity polarization measurements.
Following this validation, the PBS (PBS 4) was inserted into the final analysis setup (Figure 1.11). The beam emerging from the black fiber was collimated using a fixed $f = 18\,\mathrm{mm}$ fiber collimator and sent through an HWP (HWP 6) followed by the PBS (PBS 4). The transmitted and reflected components were detected simultaneously by two photodiodes: the transmitted component was measured with a home-made photodiode (Channel 1), and the reflected component with a Thorlabs PDA100A2 (Channel 2).

The HWP was adjusted to obtain approximately equal power at the two PBS outputs, corresponding to an input polarization close to 45° relative to the PBS axes. At this operating point, a small polarization rotation transfers power between the two outputs, producing anticorrelated variations in the two detector signals and maximizing the sensitivity to polarization fluctuations.

1.5.2 Alignment to the fiber principal axes

To ensure a stable measurement, the polarization analyzer was referenced to the principal axes of the polarization-maintaining (PM) fiber (Fig. 1.14). This required an input HWP before the fiber injection. On the output side, the beam passed through the collimator, a steering mirror, the second HWP, the characterized PBS, and a power meter.
The two HWPs were iteratively adjusted to maximize the power transmitted through the PBS while minimizing the reflected power and its temporal fluctuations. This condition ensures that only one principal axis of the PM fiber is excited and that the emerging linear polarization is perfectly aligned with one PBS axis. The optimal input-HWP angle was found to be 131°.

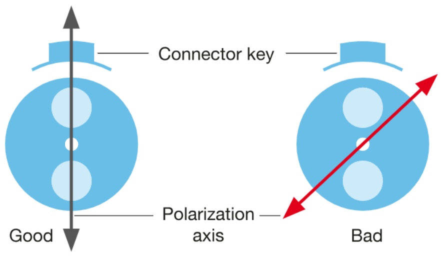


Figure 1.14: *Orientation of the input linear polarization relative to the slow and fast axes of the polarization-maintaining fiber. Adapted from [4].*

1.5.3 Photodiode calibration and data rescaling

Data were acquired using a Rohde & Schwarz RTM3K-24 oscilloscope. Since oscilloscopes record voltage traces, the raw data from both channels had to be converted into equivalent optical power (required for the Allan deviation analysis) and actual photocurrent (required to compute the theoretical shot-noise limit).
To ensure both measurements were represented on the same physical scale, the photodiode responsivity $\mathcal{R}$ (in A/W) and the transimpedance gain G_{TIA} (in V/A, physically equivalent to Ω) of each detector were used to perform the data conversion.
The home-made detector comprises a BPW34 photodiode coupled to an OP27G operational amplifier acting as a transimpedance amplifier (TIA) with a fixed gain of $G_{\mathrm{TIA}} = 10\,\mathrm{k}\Omega$ (i.e., $10^4\,\mathrm{V/A}$). The effective photodiode responsivity at 1064 nm is estimated to

be

$$\mathcal{R}_{\mathrm{HM}} \approx 0.7\,\mathrm{A/W} \times 0.3 = 0.21\,\mathrm{A/W}. \tag{1.51}$$

Consequently, its overall voltage responsivity is

$$G_{\mathrm{HM}} = \mathcal{R}_{\mathrm{HM}} G_{\mathrm{TIA}} \approx 2.1 \times 10^3\,\mathrm{V/W} = 2.1\,\mathrm{V/mW}. \tag{1.52}$$

For the Thorlabs PDA100A2 detector, assuming a standard photodiode responsivity of $\mathcal{R}_{\mathrm{TH}} = 0.4\,\mathrm{A/W}$ at 1064 nm, the data were rescaled according to the specific hardware transimpedance gain applied during the respective measurement (e.g., $1.5 \times 10^3\,\mathrm{V/A}$ at 0 dB, or $4.75 \times 10^3\,\mathrm{V/A}$ at 10 dB). Any additional electrical attenuation was also accounted for to correctly map the recorded voltages back to the incident optical power.

1.5.4 Fractional stability and differential signal

For each measurement, 10^5 samples were acquired simultaneously from both channels. Traces of 600 ms, 60 s, and 600 s total duration were recorded to probe different timescales; the 60 s trace was acquired both in standard and High-Resolution sampling modes to improve the resolution of slower fluctuations. The sampling interval τ_0 and the corresponding sampling frequency $f_s = 1/\tau_0$ were extracted directly from the recorded time vector.

After calibration, the physical power traces were used to evaluate both common-mode and differential fluctuations. To evaluate the power stability on a single channel, the signal was normalized to compute the fractional power fluctuations:

$$y(t) = \frac{P(t) - \overline{P}}{\overline{P}}, \tag{1.53}$$

where $\overline{P}$ is the mean optical power. Similarly, to isolate polarization stability, the normalized differential signal between the transmitted (P_{T}) and reflected (P_{R}) powers was analyzed:

$$D(t) = \frac{P_{\mathrm{T}}(t) - P_{\mathrm{R}}(t)}{P_{\mathrm{T}}(t) + P_{\mathrm{R}}(t)}. \tag{1.54}$$

Because the HWP was set to balance the two outputs, a fluctuation of the total optical power appears mainly in the denominator (sum), while a pure polarization fluctuation transfers power between the two PBS ports, appearing predominantly in the numerator of $D(t)$. Assuming the polarization state remains linear, the instantaneous polarization angle $\theta(t)$ can be retrieved by inverting Malus's law:

$$\theta(t) = \frac{1}{2}\arccos(D(t)). \tag{1.55}$$

The time series of the zero-centered angular fluctuation, $\Delta\theta(t) = \theta(t) - \overline{\theta}$, is then analyzed using the overlapping Allan deviation, denoted as $\sigma_\theta(\tau)$, to quantify the long-term polarization stability.

1.5.5 Allan-deviation analysis

Standard statistical variance is often insufficient for characterizing the long-term stability of optical systems, as it diverges in the presence of low-frequency fluctuations, such as $1/f$ noise, and non-stationary environmental drifts. Therefore, the time-domain stability of the fractional signals was quantified using the Allan deviation, a two-sample variance that reliably converges for all common noise types [5].

For a discrete fractional stability array y_k, $k = 1, \ldots, N$, acquired with a sampling interval τ_0, the data is grouped into sliding windows of duration $\tau = m\tau_0$. The average fractional fluctuation over the window starting at sample k is given by:

$$\overline{y}_k(\tau) = \frac{1}{m} \sum_{i=k}^{k+m-1} y_i, \qquad k = 1, \ldots, N - m + 1. \tag{1.56}$$

Unlike the standard (non-overlapping) estimator, in which only disjoint blocks of m samples are compared, the window start index k here is advanced by a single sample at a time, so that consecutive windows $\overline{y}_k(\tau)$ overlap by $m - 1$ samples. The Allan variance $\sigma_y^2(\tau)$ is defined as half the expected value of the squared difference between two such averages separated by one interval τ, i.e. by m samples. To maximize statistical confidence, particularly at longer integration times, the fully overlapping estimator was employed, which averages this squared difference over every available pair of m-sample windows in the record [5]:

$$\sigma_y^2(\tau) = \frac{1}{2\,(N - 2m + 1)} \sum_{k=1}^{N-2m+1} \left[\overline{y}_{k+m}(\tau) - \overline{y}_k(\tau)\right]^2, \qquad \sigma_y(\tau) = \sqrt{\sigma_y^2(\tau)}. \tag{1.57}$$

Because the window start index k runs over every sample rather than every m-th sample, this estimator makes use of all $N - 2m + 1$ available overlapping pairs at each averaging time τ, rather than the $\lfloor N/m \rfloor - 1$ pairs available to the non-overlapping estimator. This improves the confidence of the estimate at large τ, where the number of independent non-overlapping blocks would otherwise become small, at the cost of the resulting $\sigma_y(\tau)$ values at different τ no longer being statistically independent of one another [5].

Physically, the Allan deviation $\sigma_y(\tau)$ quantifies the typical fractional change in the optical signal between two observation windows of length τ separated by one interval τ. The behavior of $\sigma_y(\tau)$ as a function of the integration time reveals the specific noise processes dominating the system. In the short-term regime, where white noise prevails, $\sigma_y(\tau)$ typically decreases proportionally to $1/\sqrt{\tau}$, demonstrating that time-averaging suppresses fast random fluctuations. A global minimum in the curve identifies the optimal integration time, marking the ultimate stability limit of the setup. An increase in $\sigma_y(\tau)$ at longer timescales indicates the onset of uncompensated low-frequency instabilities, such as thermal drifts, air currents, or slow mechanical relaxations of the optical mounts.

1.5.6 Frequency-domain analysis and shot-noise limit

The frequency content of the power fluctuations was analyzed by computing the Amplitude Spectral Density (ASD) of the calibrated photocurrent. The mean value was subtracted from the trace, and a direct Fast Fourier Transform (FFT) was computed. The ASD describes how the amplitude of the current fluctuations is distributed over frequency, in units of $\mathrm{A}/\sqrt{\mathrm{Hz}}$.

To correctly evaluate the technical noise of the laser setup, the light-power fluctuations must be extracted from the measured photocurrent fluctuations by accounting for the quantum efficiency of the detector and separating the fundamental shot noise from the excess technical noise. The conversion of a photon flux into an electron flux is a statistical process. In a given time interval Δt, the variance of the generated electron number, σ_n^2, is related to the variance of the incident photon number, σ_N^2, by the fundamental relation [6]:

$$\sigma_n^2 = (\mathrm{QE})^2 \sigma_N^2 + \bar{n}, \tag{1.58}$$

where QE is the quantum efficiency of the photodiode, and $\bar{n} = \mathrm{QE}\bar{N}$ is the average number of generated electrons in Δt ($\bar{N}$ being the average number of incident photons). The linear term $\bar{n}$ represents the fundamental Poissonian shot noise introduced by the detection process itself.

These discrete variables can be converted into continuous macroscopic quantities: the photocurrent $I = ne/\Delta t$ (where e is the elementary charge) and the optical power $P = \hbar\omega N/\Delta t$ (where $\hbar\omega$ is the energy of a single photon). By multiplying Equation 1.58 by $(e/\Delta t)^2$, the relation can be expressed in terms of photocurrent variance σ_I^2:

$$\sigma_I^2 = \left(\mathrm{QE}\frac{e}{\Delta t}\right)^2 \sigma_N^2 + \bar{I}\frac{e}{\Delta t}. \tag{1.59}$$

The second term on the right-hand side is the photocurrent shot-noise variance, $\sigma_{I,\mathrm{SN}}^2 = \bar{I}e/\Delta t$, which in the frequency domain (with bandwidth $\Delta f = 1/(2\Delta t)$) corresponds to a single-sided amplitude spectral density of $I_{\mathrm{SN}} = \sqrt{2e\bar{I}}$.

By isolating the photon-number variance and converting it to optical power variance σ_P^2, the relationship becomes:

$$\sigma_I^2 - \sigma_{I,\mathrm{SN}}^2 = \left(\mathrm{QE}\frac{e}{\hbar\omega}\right)^2 \sigma_P^2. \tag{1.60}$$

The term in parentheses represents the responsivity of the photodiode, defined as $\eta = \mathrm{QE}e/\hbar\omega$. For the Thorlabs PDA100A2 detector at 1064 nm, the responsivity is $\eta =$

0.4 A/W.

This theoretical white-noise floor is plotted alongside the experimental FFT magnitude (scaled by $\tau_0/\sqrt{N}$) to evaluate system performance (Sec. 3.3.2). A measured spectrum lying above this baseline indicates the presence of excess technical noise in the setup, such as laser intensity noise or polarization-to-amplitude noise conversion, on top of the fundamental quantum shot noise.

1.6 Upper Breadboard Setup and Conveyor Belt Alignment

The implementation of the optical conveyor belt requires the spatial superposition of two counter-propagating optical fields: the bottom-up beam, delivered through the vacuum chamber by the Hollow-Core Photonic Crystal Fiber (HCPCF), and the top-down beam, delivered by the black fiber described in Section 1.4. The combination and alignment of these two optical paths take place on the upper breadboard, located above the main experimental chamber. Achieving a deep and reproducible conveyor-belt potential requires precise mode matching between the two counter-propagating beams and systematic control of their relative alignment along the entire shared optical path.
The experimental work on the upper breadboard was divided into three main phases: the outcoupling and conditioning of the beam emerging from the black fiber; the design and construction of dedicated imaging systems to monitor the spatial modes and the alignment of the counter-propagating beams.

1.6.1 Black-fiber outcoupling and beam conditioning

The light injected into the black fiber on the lower bench (Section 1.4.2) is guided to the upper breadboard, where it is outcoupled, conditioned, and combined with the counter-propagating HCPCF beam. The upper-breadboard branch, shown in Figure 1.15, consists of:

- the black-fiber outcoupling assembly (Black fiber output), mounted on a custom-made support;
- an $f = 19\,\mathrm{mm}$ collimating lens (Lens f19), positioned on a high-stability mount, collimating the diverging beam emerging from the fiber;
- a half-wave plate (HWP 1), used to adjust the polarization of the collimated beam, and a polarizing beam splitter (PBS 1);
- an adjustable beam expander (Beam expander) providing a magnification of approximately $3\times$ and fine adjustement of the beam divergence, matching the transverse size of the black-fiber beam to the expected size of the beam emerging from the HCPCF and changing the focal plane of the collimating lens;
- two mirrors (Mirror 1 and Mirror 2), directing the expanded beam toward the dichroic mirror;

- another set of HWP (HWP 2) and PBS (PBS 2), used to adjust the polarization of the top-down beam before it enters the vacuum chamber (Opening above the vacuum chamber), ensuring that it matches the polarization of the bottom-up beam emerging from the HCPCF;
- a dichroic mirror (Dichroic Mirror), reflecting the 1064 nm light downward, into the vacuum chamber;
- an $f = 200$ mm lens (Lens f200), mounted along the vertical axis (on a z-stage) downstream of the dichroic mirror, fulfilling a dual purpose: it focuses the top-down beam from the black fiber onto the upper tip of the HCPCF, while simultaneously collimating the counter-propagating bottom-up beam emerging from the same hollow-core fiber (Top of vacuum chamber).

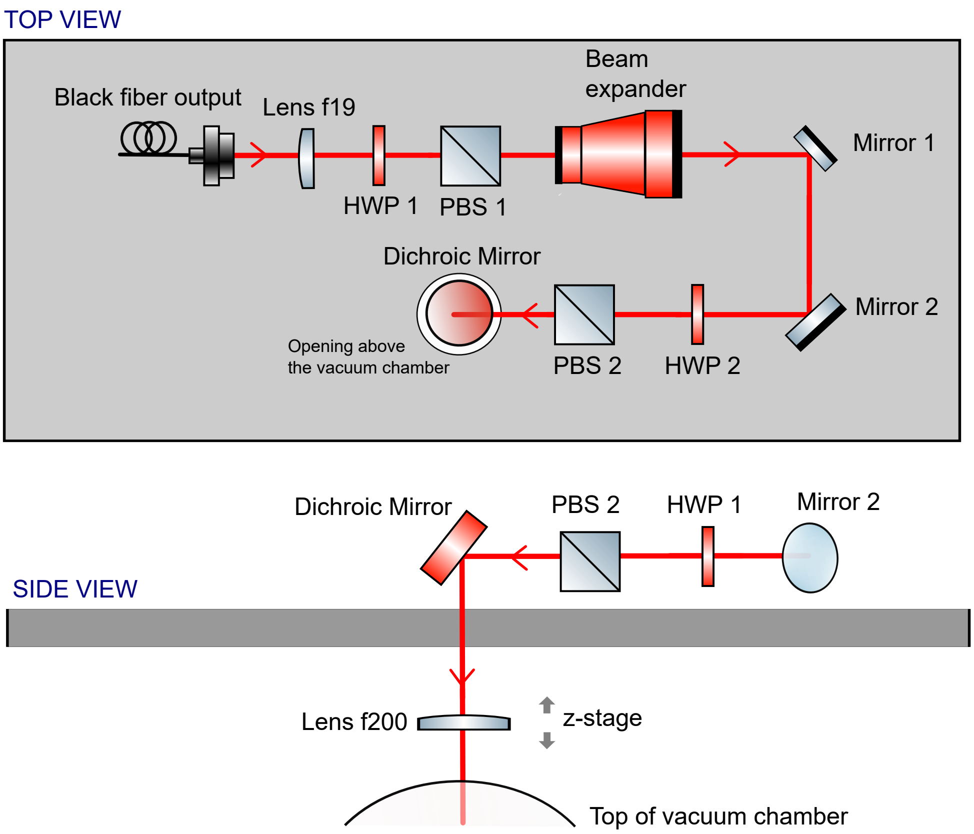


Figure 1.15: *Optical scheme of the upper-breadboard branch, where the top-down and bottom-up beams are combined above the vacuum chamber.*

The design of the shared optical path requires a careful evaluation of the beam size at each stage. The expected waist radius w_{col} of the bottom-up collimated beam can be estimated from the nominal mode radius within the hollow core, $w_{\text{HCPCF}} \approx 19\,\mu$m, using the relation

$$w_{\text{col}} = \frac{\lambda f}{\pi w_{\text{HCPCF}}}, \tag{1.61}$$

where λ is the wavelength and f is the focal length of the vertical lens. For $\lambda = 1064\,\mathrm{nm}$ and $f = 200\,\mathrm{mm}$, Equation (1.61) gives $w_{\mathrm{col}} \approx 3.6\,\mathrm{mm}$, corresponding to a full beam diameter $d_{\mathrm{col}} = 2w_{\mathrm{col}} \approx 7.2\,\mathrm{mm}$.

This collimated beam is steered by standard 1-inch (25.4 mm) mirrors. At an incidence angle of 45°, the beam footprint on the mirror surface becomes elliptical, with major axis d^* given by

$$d^* = \frac{d_{\mathrm{col}}}{\cos(45^\circ)} \approx 10.2\,\mathrm{mm}. \tag{1.62}$$

Although this footprint is mathematically smaller than the clear aperture of a 1-inch mirror, the available clearance is relatively modest. If the effective mode radius in the HCPCF is smaller than the assumed $19\,\mu\mathrm{m}$, as might be expected given the $48\,\mu\mathrm{m}$ core diameter, the collimated beam divergence and the resulting mirror footprint would be correspondingly larger. Careful alignment, verified with an infrared viewer, is therefore critical to prevent clipping and vignetting at the edges of the mirrors.

1.6.2 Mode imaging and alignment strategy

Aligning this counter-propagating configuration and ensuring optimal mode matching requires firstly simultaneous monitoring of the spatial mode of the HCPCF and optimization of the bottom-up injection. For this purpose, an imaging system (microscope) was assembled on the upper breadboard, exploiting the shared optical path. The imaging branch, shown in Figure 1.16, consists of:

- a $f = 200\,\mathrm{mm}$ lens (Lens f200) mounted on a vertical shifting z-stage (z-stage) above the vacuum chamber (Top of vacuum chamber), to collimate the 1064 nm light coming out of the top of the vacuum chamber;
- a dichroic mirror (Dichroic Mirror), reflecting the 1064 nm light;
- a magnetic flippable mirror (Magnetic mirror), inserted between the dichroic mirror and the first mirror of the black-fiber branch, picking off the bottom-up beam;
- an $f = 300\,\mathrm{mm}$ lens (Lens f300);
- Mirror 3, directing the beam toward the focusing optics;
- a Zeiss microscope objective (Objective f16.5) (16.5 mm, Plan Neofluar $10\times/0.30$);
- Mirror 4 and Mirror 5, directing the beam toward the focusing optics and the camera;
- a final $f = 500\,\mathrm{mm}$ focusing lens (Lens f500), projecting the image onto a CMOS camera (Camera).

This multi-stage magnification (~ 45) setup provides the resolution needed to resolve the spatial distribution of the light exiting the microstructured fiber.

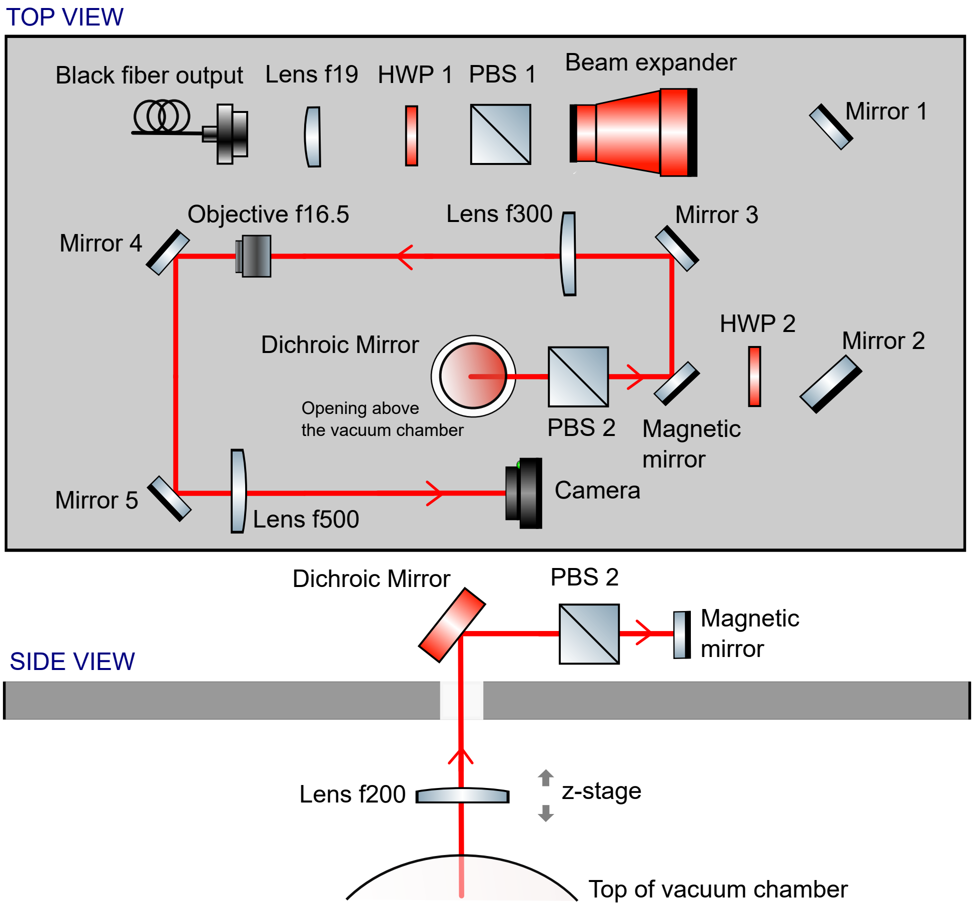


Figure 1.16: *Optical scheme of the imaging (microscope) branch used to monitor the spatial mode of the HCPCF on the upper breadboard.*

A picture of the full upper breadboard setup is shown in Figure 1.17.

The alignment method relies on the assumption that the mode in the HCPCF must be injected symmetrically and propagate accordingly. During the initial observations with the microscope, the fundamental mode was found not to propagate symmetrically within the core. By applying mechanical perturbations to the fiber, higher-order propagation modes were intentionally excited; this behavior was systematically exploited to map the spatial boundaries of the hollow core. Once the core region was identified, the fiber position was finely adjusted to ensure that the mode extended symmetrically across the entire available area. Exploiting the small transmission of the dichroic mirror at 1064 nm makes it possible to simultaneously optimize the injection (by placing a power meter directly above it) and monitor the mode in real time. This optimization yielded a symmetric fundamental Gaussian mode with an overall bottom-up injection efficiency of approximately 30%.

With the HCPCF mode optimized, the spatial overlap between the top-down and bottom-up beams was verified. An additional mirror, placed temporarily after the dichroic mirror, was used to project the beam profiles onto a distant surface and checked using a viewing

Figure 1.17: *Photograph of the complete upper breadboard setup.*

card, to verify the collimation of the HCPCF beam; the procedure was repeated for the black-fiber beam, sampling it after the beam expander and at a far-field plane. Both beams were then turned on simultaneously to verify that they were correctly centered on all lenses and mirrors and perfectly overlapped along the entire shared path.

1.6.3 Figures of merit for the injection efficiency

To quantitatively evaluate the optical setup and the alignment procedure, five figures of merit were established and continuously monitored:

1. **Bottom-up HCPCF mode quality**: the symmetry and spatial distribution of the beam emerging from the vacuum chamber, evaluated through the upper-breadboard microscope (Section 1.6.2);

2. **Bottom-up HCPCF injection efficiency**: the ratio of the power transmitted above the vacuum chamber to the incident power measured before the lower injection lens, as in Section 1.6.2;

3. **Retro-injection into the black fiber**: the coupling efficiency of the bottom-up HCPCF beam into the black fiber. Since the two beams must be perfectly collinear to form the conveyor belt, optimizing this reverse-coupling signal guarantees the spatial overlap of the two modes on the upper breadboard;

4. **Top-down injection into the HCPCF**: the coupling of the beam delivered by the black fiber into the hollow-core fiber, from above, measured by the ratio of the power collected after the lower collimating lens and before the transmission through the vacuum chamber;

5. **Top-down mode quality**: the symmetry and spatial distribution of the beam injected into the hollow-core fiber from above.

To evaluate the last figure of merit, a secondary microscope was built on the lower optical bench (Fig. 1.18), exploiting the small but non-zero transmission of the lower 1064 nm dichroic mirror. This diagnostic branch consists of:

- the $f = 50\,\mathrm{mm}$ lens already present for the injection of the HC branch;

- a flippable mirror, picking off the light exiting the bottom tip of the HCPCF(mounted on a flip base to allow alternating operation with the 780 nm injection branch, not used in this work);

- an additional mirror(Mirror 2), directing the beam toward the focusing optics;

- an $f = 1000\,\mathrm{mm}$ focusing lens, projecting the beam onto a camera.

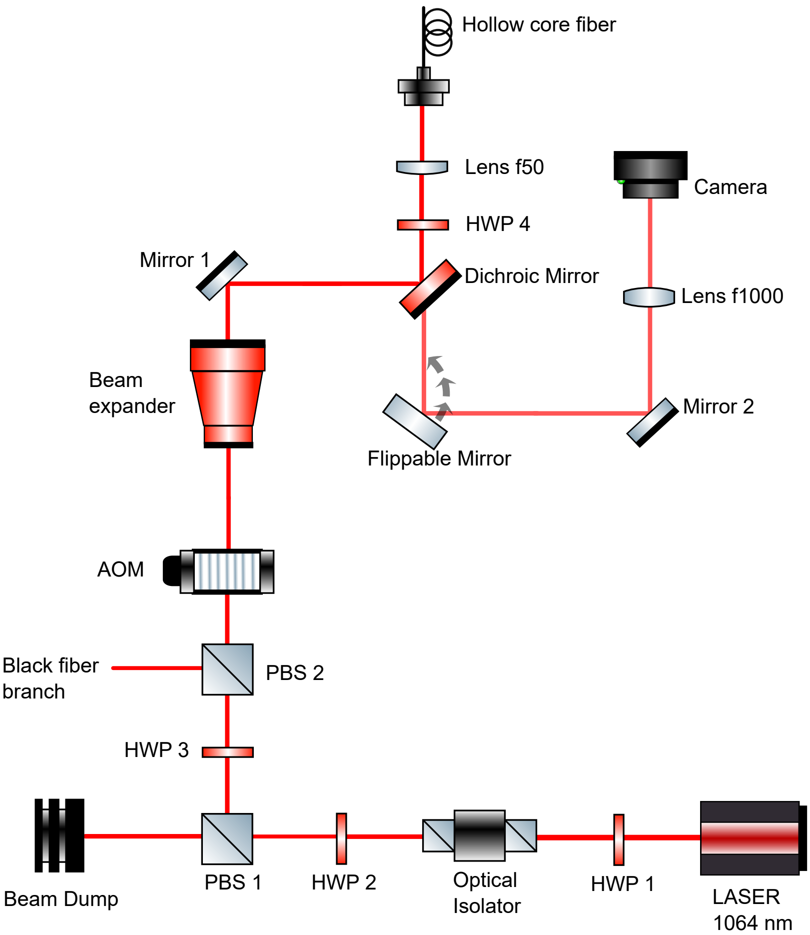
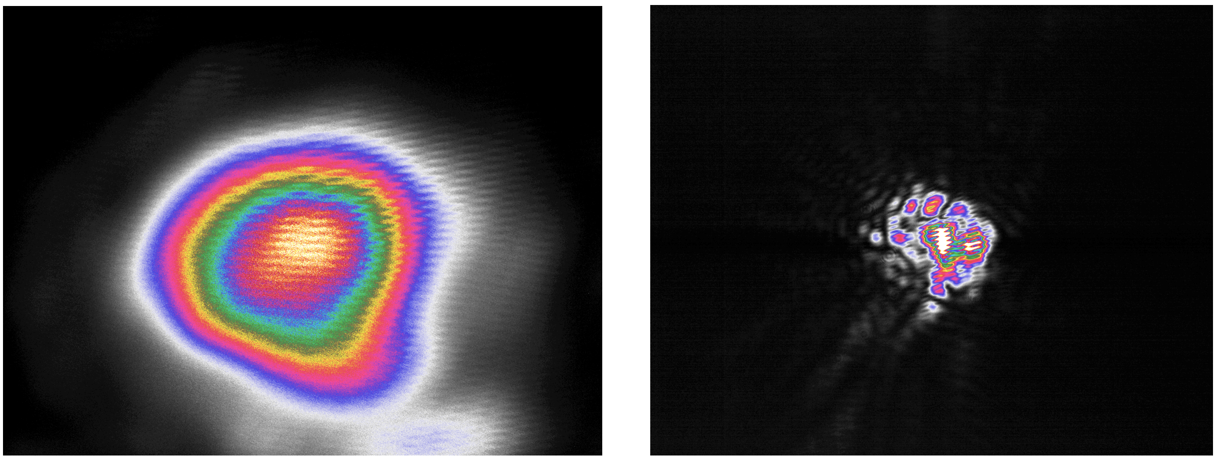


Figure 1.18: *Schematic of the secondary microscope diagnostic branch on the lower optical bench.*

This diagnostic tool allows the simultaneous optimization of the top-down injection and the real-time monitoring of the spatial mode traversing the vacuum chamber from top to bottom.

Ultimately, the polarization across the setup was optimized: for the upper breadboard and the black fiber, this was achieved using the black fiber test bench method which relies on the presence of the two half wave plates and polarizing beam splitters (described in Sec. 1.5.2), while the waveplate on the lower bench was employed to minimimize the reflected component of PBS2 on the upper breadboard and maximize retro-injection. The macroscopic alignment of the beams was successfully achieved by controlling their collimation and utilizing the beam expander, while fine spatial control relied on the upper mirrors, the fiber injection optics, and the shared $f = 200\,\mathrm{mm}$ lens.

Following this systematic optimization,a high bottom-up mode quality was achieved, as verified through the upper microscope (Fig. 1.19a), and a bottom-up injection efficiency of 32%. The retro-injection efficiency into the black fiber reached 60% (compared to a baseline direct injection efficiency of 75%). Furthermore, a final top-down injection efficiency of 35% was measured. Evaluating the corresponding spatial profile with the lower microscope (Fig. 1.19b) revealed that the top-down mode is not perfectly single-mode. This deviation from an ideal fundamental mode can be attributed both to the potential for further improvements in the optical alignment and diagnostic procedures, and to a possible deterioration of the fiber tip located inside the vacuum chamber.



(a) (b)

Figure 1.19: *Pseudocolored intensity images of the spatial modes. (a) Bottom-up beam mode detected by the upper microscope. (b) Top-down beam mode detected by the lower microscope.*

1.6.4 RF electronics characterization for intensity control

The depth of the optical dipole trap and the dynamics of the conveyor belt require precise control over the optical power of both beams. The optical power in both the main ODT branch and the black-fiber branch is actively controlled using Acousto-Optic Modulators (AOMs), whose Radio-Frequency (RF) drive signals are generated by a homemade Direct Digital Synthesizer (DDS) board, designated DDS3. The RF signals are amplified by two power amplifiers and then delivered to the AOMs.
To guarantee optimal performance, the first-order transmission profile of the black-fiber AOM was verified to ensure it operates at high efficiency (Figure 1.20). This procedure follows the characterization previously performed for the HC branch AOM, which operates at an optimal frequency of 110 MHz and an amplitude setting of 420. Establishing this high-efficiency is a crucial step for the subsequent automatic implementation of the conveyor belt.

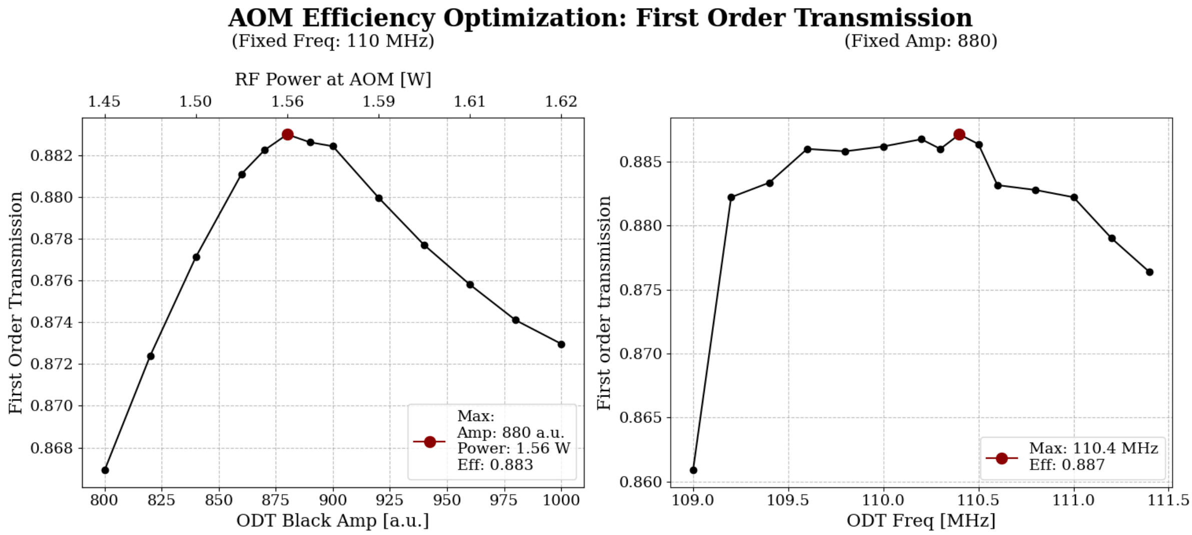


Figure 1.20: *First-order transmission profile of the black-fiber branch AOM. The left panel shows the transmission as a function of the DDS amplitude and the corresponding power reaching the AOM after the power amplifier (at a fixed frequency of 110 MHz). The right panel shows the transmission as a function of the RF frequency (at a fixed amplitude of 880).*

Although the characterization of the black-fiber AOM reveals a peak transmission at 110.4 MHz, the first order transmission profile is sufficiently broad to allow operation at the standard frequency of 110 MHz used for the HCPCF branch.

1.7 Numerical Model of EIT Cooling

Once the atoms enter the fiber, they must be efficiently cooled to control their motion (Sec. 1.4). Simulating this process is a necessary step before modeling the atoms' behavior in the full experimental system. The EIT cooling dynamics are simulated using QuTiP [7], solving the open-system evolution of an atom coupled to a quantized motional mode. Two models of increasing complexity are implemented. The first is an ideal three-level Λ system, used to validate the EIT cooling mechanism and the numerical implementation against the analytical predictions of Sec. 1.4. The second includes the complete hyperfine and Zeeman structure of the ^{87}Rb D_2 line, and is used to assess off-resonant excitation, spontaneous emission, optical pumping, and repumping.

Both descriptions are formulated within the same Lindblad master-equation framework and differ only in the dimension and structure of the internal atomic Hilbert space. The motion is treated in one spatial dimension throughout; the simulated direction is set by projecting each laser's wavevector onto the corresponding trap axis.

1.7.1 Internal Atomic Basis

In the reduced model, the internal Hilbert space is spanned by two ground states, $|g_1\rangle$ and $|g_2\rangle$, and one excited state, $|e\rangle$. A weak probe field with Rabi frequency Ω_p couples $|g_1\rangle$ to $|e\rangle$, while a stronger coupling field with Rabi frequency Ω_c couples $|g_2\rangle$ to $|e\rangle$. At two-photon resonance this model supports the dark state of Eq. (1.59) and reproduces the ideal carrier suppression discussed in Sec. 1.4. The three-level calculation is used to test the Hamiltonian construction, the motional coupling, the collapse operators, and the time-evolution routines before introducing the complete atomic structure.

The multilevel model includes all hyperfine and Zeeman sublevels of the $5S_{1/2}$ and $5P_{3/2}$ manifolds of ^{87}Rb. The internal basis is written as $|F, m_F\rangle$ and comprises the ground manifolds

$$F = 1, \qquad F = 2, \tag{1.63}$$

with three and five Zeeman states respectively, and the excited manifolds

$$F' = 0, 1, 2, 3, \tag{1.64}$$

containing sixteen states in total. The dimension of the internal Hilbert space is therefore

$$\dim(\mathcal{H}_{\rm at}) = 3 + 5 + 16 = 24. \tag{1.65}$$

All dipole-allowed couplings are constructed explicitly. For a transition between a ground state $|F_g, m_g\rangle$ and an excited state $|F_e, m_e\rangle$, driven by a spherical polarization component

$q \in \{-1, 0, +1\}$, the coupling strength is calculated from the hyperfine dipole matrix element [8],

$$d_{eg}^{(q)} = \langle F_e, m_e | d_q | F_g, m_g \rangle = \langle F_e \| d \| F_g \rangle (-1)^{F_e - m_e} \begin{pmatrix} F_e & 1 & F_g \\ -m_e & q & m_g \end{pmatrix}, \tag{1.66}$$

where the $3j$ symbol carries the electric-dipole selection rules, and the reduced hyperfine matrix element $\langle F_e \| d \| F_g \rangle$ sets the relative strength between transitions connecting different hyperfine manifolds, through

$$\langle F_e \| d \| F_g \rangle = \langle J_e \| d \| J_g \rangle (-1)^{F_e + J_g + 1 + I} \sqrt{(2F_e + 1)(2J_g + 1)} \begin{Bmatrix} J_g & J_e & 1 \\ F_e & F_g & I \end{Bmatrix}, \tag{1.67}$$

with $J_g = 1/2$, $J_e = 3/2$, and $I = 3/2$ for the ^{87}Rb D_2 line. Transitions that violate the electric-dipole selection rules are assigned zero coupling.

To reduce optical pumping into states outside the EIT cycle, the target Λ system is built around states near the edge of a Zeeman manifold rather than around $m_F = 0$: the condition $\Delta m_F = 0, \pm 1$ then limits the number of ground states reachable by spontaneous decay. In the configuration used for the multilevel and axial-parameter simulations (Fig. 1.21), the two EIT ground states belong to different hyperfine manifolds. The σ^--polarized probe field addresses $|g_1\rangle = |F = 2, m_F = -1\rangle \to |F' = 2, m_F = -2\rangle$ and is scanned across the two-photon resonance. The σ^--polarized coupling field addresses $|g_2\rangle = |F = 1, m_F = -1\rangle \to |F' = 2, m_F = -2\rangle$; its Rabi frequency is set by the EIT cooling condition of Eq. (1.100). A π-polarized repump field, resonant on $|F = 2, m_F = -2\rangle \to |F' = 2, m_F = -2\rangle$, recovers population that leaks out of the Λ system into the adjacent Zeeman sublevel of $F = 2$; its motional factor is set to the identity, Eq. (1.81). This configuration is used for the simulation with the axial trap parameters (Sec. 1.7.11).

1.7.2 Motional Hilbert Space

The atomic motion along the selected trap axis is approximated by a harmonic oscillator of angular frequency ν. The total Hilbert space is

$$\mathcal{H} = \mathcal{H}_{\text{at}} \otimes \mathcal{H}_{\text{mot}}, \tag{1.68}$$

where

$$\mathcal{H}_{\text{mot}} = \text{span}\{|0\rangle, \ldots, |N_{\text{vib}} - 1\rangle\}. \tag{1.69}$$

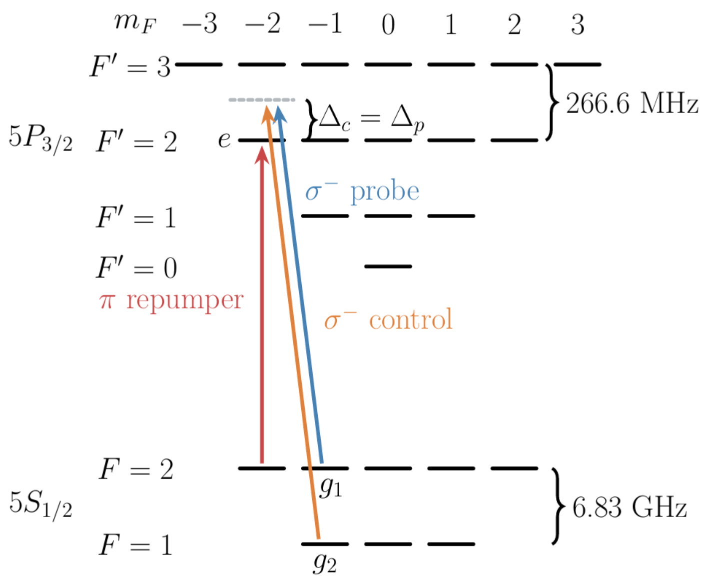


Figure 1.21: *Level scheme of the* ^{87}Rb D_2 *transition implemented in the multilevel numerical model. The EIT* Λ *system is formed by the ground states* $|g_1\rangle = |F = 2, m_F = -1\rangle$ *and* $|g_2\rangle = |F = 1, m_F = -1\rangle$, *both coupled to the excited state* $|e\rangle = |F' = 2, m_F = -2\rangle$.

The motional Hamiltonian is

$$H_{\text{mot}} = \hbar\nu \left(a^\dagger a + \frac{1}{2}\right), \tag{1.70}$$

with a and $a^\dagger$ the annihilation and creation operators. The truncation N_{vib} is chosen by balancing numerical cost against convergence of the motional distribution.

1.7.3 Laser Coupling and Lamb–Dicke Expansion

For a laser field l with wavevector $\mathbf{k}_l$, optical phase ϕ_l, and polarization components $\epsilon_{l,q}$, the position-dependent rotating-wave interaction is written as

$$H_l = \frac{\hbar}{2} \sum_{g,e,q} \Omega_l \epsilon_{l,q} \mathcal{C}_{eg}^{(q)} |e\rangle\langle g| \, e^{ik_{l,\parallel}x + i\phi_l} + \text{h.c.}, \tag{1.71}$$

where $\mathcal{C}_{eg}^{(q)} = d_{eg}^{(q)}/d_{\text{target}}$ is the dipole matrix element of Eq. (1.66), normalized to the value d_{target} of the transition each beam is designed to drive, and

$$k_{l,\parallel} = \mathbf{k}_l \cdot \hat{\mathbf{e}}_{\text{mot}} \tag{1.72}$$

is the projection of the laser wavevector along the simulated motional axis.

Using

$$x = x_0(a + a^\dagger), \qquad x_0 = \sqrt{\frac{\hbar}{2m\nu}}, \tag{1.73}$$

the Lamb–Dicke parameter associated with field l is

$$\eta_l = k_{l,\parallel} x_0. \tag{1.74}$$

In the Lamb–Dicke regime, the position-dependent phase is expanded to first order,

$$e^{ik_{l,\parallel}x} \simeq 1 + i\eta_l(a + a^\dagger). \tag{1.75}$$

This expansion retains the carrier and first motional sidebands while neglecting higher-order processes.

For a Raman process involving absorption from one field and stimulated emission into another, the two-photon momentum transfer is set by

$$\Delta\mathbf{k} = \mathbf{k}_p - \mathbf{k}_c, \tag{1.76}$$

consistent with Sec. 1.4. The corresponding effective Lamb–Dicke parameter along the simulated direction is

$$\eta_{\text{eff}} = (\mathbf{k}_p - \mathbf{k}_c) \cdot \hat{\mathbf{e}}_{\text{mot}}\, x_0. \tag{1.77}$$

In the three-level benchmark, the coupling beam propagates orthogonally to the motional axis, so $\eta_c = 0$, while the probe field carries the motional coupling. In the multilevel model the framework allows an independent value of η_l for each beam, according to the experimental geometry.

1.7.4 Hamiltonian in the Rotating Frame

After transformation to a rotating frame and application of the rotating-wave approximation, the Hamiltonian is

$$H = H_{\text{at}} + H_{\text{mot}} + H_{\text{Z}} + \sum_l H_l. \tag{1.78}$$

Here H_{at} contains the hyperfine energies and laser detunings in the selected rotating frame, and

$$H_{\text{Z}} = \sum_i g_{F_i} \mu_B B m_{F_i} |i\rangle\langle i| \otimes \mathbb{I}_{\text{mot}} \tag{1.79}$$

describes the linear Zeeman shift produced by the quantization field B, assumed weak enough that F and m_F remain good labels.

For the three-level system, the implemented Hamiltonian is

$$\begin{aligned} H_\Lambda = {} & \hbar\Delta_p|g_1\rangle\langle g_1| + \hbar\Delta_c|g_2\rangle\langle g_2| + \hbar\nu a^\dagger a \\ & + \frac{\hbar\Omega_c}{2}\left(|e\rangle\langle g_2| + |g_2\rangle\langle e|\right) \\ & + \frac{\hbar\Omega_p}{2}\left[|e\rangle\langle g_1|\left(1 + i\eta_p(a + a^\dagger)\right) + \text{h.c.}\right]. \end{aligned} \tag{1.80}$$

In the multilevel model, Eq. (1.71) is constructed separately for the probe, coupling, and repump fields, each with its own Rabi frequency, detuning, polarization, and projected wavevector. The repump field returns population from ground states outside the intended Λ subsystem. Its motional factor is set to the identity,

$$e^{ik_{\text{rep},\parallel}x} \to \mathbb{I}_{\text{mot}}, \tag{1.81}$$

so repump-induced motional sidebands are neglected. This reduces the numerical complexity and isolates the repumper's role in recycling internal population.

1.7.5 Spontaneous Emission and Master Equation

The density operator evolves according to

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k\rho L_k^\dagger - \frac{1}{2}L_k^\dagger L_k\rho - \frac{1}{2}\rho L_k^\dagger L_k\right). \tag{1.82}$$

For the three-level system, symmetric branching into the two ground states is assumed:

$$L_1 = \sqrt{\frac{\gamma}{2}}\,|g_1\rangle\langle e| \otimes \mathbb{I}_{\text{mot}}, \qquad L_2 = \sqrt{\frac{\gamma}{2}}\,|g_2\rangle\langle e| \otimes \mathbb{I}_{\text{mot}}. \tag{1.83}$$

For the multilevel system, the collapse operator for a decay channel $e \to g$ is

$$L_{e\to g} = \sqrt{\gamma_e b_{eg}}\,|g\rangle\langle e| \otimes \mathbb{I}_{\text{mot}}, \tag{1.84}$$

where γ_e is the total decay rate of the excited state and b_{eg} the normalized branching fraction,

$$b_{eg} = \frac{\sum_q \left|d_{eg}^{(q)}\right|^2}{\sum_{g',q}\left|d_{eg'}^{(q)}\right|^2}, \qquad \sum_g b_{eg} = 1, \tag{1.85}$$

so that, in general,

$$\sum_g L_{e\to g}^\dagger L_{e\to g} = \gamma_e|e\rangle\langle e|. \tag{1.86}$$

With the full hyperfine-weighted dipole matrix elements of Eq. (1.66), b_{eg} is normalized per excited sublevel by construction, Eq. (1.85), so the identity above holds exactly, with $\gamma_e = \gamma$ for every excited hyperfine sublevel. The total decay rate is therefore independent of F', consistent with the physical picture of a single natural linewidth for the D_2 line.

The model accounts for motional changes driven by laser absorption and stimulated coupling, but not for recoil heating during spontaneous emission. Within the Lamb–Dicke regime this correction enters at order η^2; it can still affect the final occupation and should be included for a quantitative comparison with the theoretical limit of Eq. (1.74) or with experimental temperatures.

1.7.6 Steady-State Spectroscopy

The EIT spectrum is calculated by scanning the probe detuning while holding the remaining laser parameters fixed. At each detuning, the stationary density operator is obtained from

$$\mathcal{L}_{\mathrm{tot}}(\rho_{\mathrm{ss}}) = 0, \qquad \mathrm{Tr}(\rho_{\mathrm{ss}}) = 1, \tag{1.87}$$

where $\mathcal{L}_{\mathrm{tot}}$ is the full Liouvillian superoperator.

The main spectral observable is the total excited-state population,

$$P_e(\Delta_p) = \mathrm{Tr}\left[\rho_{\mathrm{ss}}(\Delta_p) \sum_e |e\rangle\langle e|\right]. \tag{1.88}$$

In the weak-probe regime this is proportional to the total photon-scattering rate and is used as an absorption proxy.

For the multilevel calculation, additional projectors distinguish excitation from population leakage:

$$P_{F'=2} = \sum_{e\in F'=2} |e\rangle\langle e|, \tag{1.89}$$

$$P_{F'=3} = \sum_{e\in F'=3} |e\rangle\langle e|, \tag{1.90}$$

$$P_{\mathrm{out}} = \sum_{g\notin \mathcal{G}_\Lambda} |g\rangle\langle g|, \tag{1.91}$$

where $\mathcal{G}_\Lambda$ is the pair of ground states defining the target Λ system. These observables identify the dark resonance, quantify off-resonant excitation, and track population accumulated in ground states not directly addressed by the EIT beams.

1.7.7 Time-Dependent Cooling Dynamics

Cooling is quantified by propagating an initial atom–motion state and evaluating the phonon number expectation value,

$$\langle n(t)\rangle = \mathrm{Tr}\left[\rho(t)a^\dagger a\right]. \tag{1.92}$$

The initial internal state is chosen within the active ground-state manifold. The motional state is specified either as a Fock state or as a thermal mixture,

$$\rho_{\mathrm{th}} = \sum_{n=0}^{N_{\mathrm{vib}}-1} p_n|n\rangle\langle n|, \qquad p_n = \frac{\bar{n}_0^n}{(1+\bar{n}_0)^{n+1}}, \tag{1.93}$$

with the probabilities renormalized after truncation when necessary.

The three-level model is propagated with QuTiP's deterministic master-equation solver `mesolve`, which evolves the density operator directly and gives noise-free ensemble expectation values. For the full multilevel model, the Monte Carlo wavefunction solver `mcsolve` is used instead [7]: it propagates stochastic pure-state trajectories interrupted by quantum jumps, estimating an observable as

$$\langle O(t)\rangle \simeq \frac{1}{N_{\mathrm{traj}}} \sum_{j=1}^{N_{\mathrm{traj}}} \langle\psi_j(t)|O|\psi_j(t)\rangle. \tag{1.94}$$

The statistical uncertainty decreases approximately as $N_{\mathrm{traj}}^{-1/2}$.

For internal dimension N_{at} and motional cutoff N_{vib}, the state-vector dimension is

$$N = N_{\mathrm{at}}N_{\mathrm{vib}}. \tag{1.95}$$

A density matrix has N^2 complex entries, a Monte Carlo trajectory a state vector with N entries; the trajectory method substantially reduces memory use for the 24-level calculation.

1.7.8 Units and Conversion to Physical Time

The numerical calculations use the natural angular linewidth as the unit of frequency,

$$\gamma = 1, \tag{1.96}$$

with $\hbar = 1$. All detunings, Rabi frequencies, hyperfine splittings, Zeeman shifts, and trap frequencies are divided by γ, and time is expressed in units of γ^{-1}.

For the ^{87}Rb D_2 transition, the linewidth is

$$\frac{\gamma}{2\pi} = 6.067 \text{ MHz}, \tag{1.97}$$

so one simulation time unit corresponds to

$$\gamma^{-1} = \frac{1}{2\pi \times 6.067 \text{ MHz}} \simeq 26.2 \text{ ns}. \tag{1.98}$$

A dimensionless interval t_{sim} converts to physical time through

$$t_{\text{phys}} = \frac{t_{\text{sim}}}{\gamma}. \tag{1.99}$$

The same angular-frequency convention is used for ν, Ω_l, Δ_l, and the Zeeman shifts, avoiding factors of 2π inside the Hamiltonian.

1.7.9 Validation of the Three-Level Model

The reduced model is validated in two stages. First, the motional degree of freedom is omitted and the probe detuning is scanned to verify the EIT transparency minimum at two-photon resonance and the narrow dressed-state resonance displaced by the coupling-induced AC Stark shift of Eq. (1.66).

Second, the motional basis is restored and the coupling intensity is set by the EIT cooling condition. For the sign convention and blue-detuned configuration of Sec. 1.4, the exact relation is Eq. (1.68),

$$\Omega_c^2 = 4\nu(\nu + \Delta_c), \tag{1.100}$$

which reduces to

$$\Omega_c^2 \simeq 4\Delta_c\nu \tag{1.101}$$

when $\nu \ll \Delta_c$. Eq. (1.100) sets Ω_c in both the three-level and multilevel codes. Under this condition, the narrow absorption resonance aligns with the red sideband while carrier excitation is suppressed by the dark resonance.

1.7.10 Optimization of the Multilevel Configuration

The multilevel laser parameters are optimized on steady-state spectra before running the more expensive time-dependent simulations. For each parameter set, the probe detuning is scanned and the following are recorded: the total excited-state population, the populations of the target and off-resonant excited manifolds, and the population outside the active Λ subsystem.

The scans proceed sequentially:

1. The repump Rabi frequency is varied to find the minimum intensity that prevents long-lived population accumulation outside the EIT cycle. Excessive repump power is avoided, since off-resonant coupling can shift and broaden the EIT spectrum.

2. The probe Rabi frequency is varied within the weak-probe regime. Increasing Ω_p increases the scattering signal and may increase the cooling rate, but it also saturates and power-broadens the dark resonance, increasing carrier scattering.

3. The coupling detuning and Rabi frequency are adjusted to align the narrow absorption maximum with the red sideband.

The optimization criterion is not the height of the absorption peak alone: parameter sets are chosen for low carrier excitation, enhanced scattering at the cooling sideband, reduced scattering at the heating sideband, and limited population leakage. The selected parameters are then tested through a time-dependent `mcsolve` calculation of $\langle n(t) \rangle$.

1.7.11 Simulation with Axial Experimental Parameters

The configuration in which the two EIT ground states belong to different hyperfine manifolds is evaluated using the axial trap parameters of the experimental system. As derived in Section 1.3, the local trapping frequencies depend on the geometric scaling of the trap parameters between the MOT loading position and the fiber tip.
At the MOT position, the trapping beams have a waist of $w_{\mathrm{MOT}} = 126\,\mu\mathrm{m}$, giving an initial interference potential depth of $U_{\mathrm{MOT}} \approx 25\,\mu\mathrm{K}$. From Eq. (1.33), the axial frequency depends on the potential depth and the trapping wavelength $\lambda = 1064\,\mathrm{nm}$:

$$\nu_{z,\mathrm{MOT}} = \frac{1}{\lambda}\sqrt{\frac{2U_{\mathrm{MOT}}}{m}}. \tag{1.102}$$

This gives $\nu_{z,\mathrm{MOT}} \approx 65\,\mathrm{kHz}$.
During transport, the atoms are guided toward the fiber tip, where the beam waist is focused down to $w_0 = 19\,\mu\mathrm{m}$. The optical potential depth scales as $U \propto 1/w^2(z)$, so the compression amplifies it by

$$\frac{U_{\mathrm{tip}}}{U_{\mathrm{MOT}}} = \left(\frac{w_{\mathrm{MOT}}}{w_0}\right)^2 = \left(\frac{126, \mu\mathrm{m}}{19, \mu\mathrm{m}}\right)^2 \approx 44, \tag{1.103}$$

bringing the trap depth at the fiber tip to $U_{\mathrm{tip}} \approx 1.1\,\mathrm{mK}$.
Since the axial frequency scales as $\nu_z \propto \sqrt{U}$, its value at the fiber tip increases by $\sqrt{44}$:

$$\nu_{z,\mathrm{tip}} = \nu_{z,\mathrm{MOT}}\sqrt{44} \approx 430, \mathrm{kHz}. \tag{1.104}$$

The radial frequency ν_r depends on both the potential depth and the local waist, scales more steeply, and reaches approximately 5 kHz.

If transport along the optical conveyor belt is adiabatic, the axial action of the trapped atoms is conserved. From an initial temperature $T_{\text{MOT}} = 8.5\,\mu\text{K}$ at the loading position, the initial mean axial phonon number follows Bose–Einstein statistics:

$$\langle n_z \rangle = \frac{1}{\exp\left(\frac{h\nu_{z,\text{MOT}}}{k_B T_{\text{MOT}}}\right) - 1} \approx 2.3. \tag{1.105}$$

During adiabatic compression, temperature scales with trapping frequency ($T \propto \nu_z$), so the axial temperature at the fiber tip increases by the same factor:

$$T_{\text{tip}} = T_{\text{MOT}}\sqrt{44} \approx 56, \mu\text{K}. \tag{1.106}$$

Despite this rise in temperature, adiabaticity keeps the mean phonon number fixed at $\langle n_z \rangle \approx 2.3$. Since this calculated value is an estimate based on the initial measured temperature, it is rounded to the nearest integer and set the initial mean phonon number to $\langle n_z \rangle = 2$ in the simulation.

The axial trap frequency $\nu_{z,\text{tip}}$ is inserted directly into the motional Hamiltonian, with ν set to 430 kHz. The Lamb–Dicke parameters of the probe and coupling beams follow from their wavevector projections, Eq. (1.74). For EIT cooling at the 780 nm absorption wavelength, the optical wavenumber is $k = 2\pi/\lambda_{\text{EIT}} \approx 8.055 \times 10^6\,\text{m}^{-1}$, and the spatial extent of the axial ground-state wavepacket is

$$x_{0,ax} = \sqrt{\frac{\hbar}{2m(2\pi\nu_{z,\text{tip}})}} \approx 11, \text{nm}. \tag{1.107}$$

The corresponding axial Lamb–Dicke parameter is

$$\eta_{ax} = kx_{0,ax} \approx 0.09. \tag{1.108}$$

Since $\eta_{ax} \ll 1$, the axial motion is within the Lamb–Dicke regime assumed in Section 1.7.3. The magnetic field, laser polarizations, hyperfine splittings, detunings, and Rabi frequencies are retained explicitly in the 24-level Hamiltonian.

The steady-state probe scan is carried out for the axial parameters, using the probe/coupling/repump assignment of Fig. 1.21. The probe, coupling, and repump parameters are adjusted until the absorption maximum aligns with the axial red sideband while the carrier stays close to the transparency minimum. The resulting parameter set is used in the Monte Carlo time evolution.

Atoms near the fiber surface experience spatially varying tensor light shifts, local magnetic field gradients, and polarization constraints that alter the internal energy structure.

Future simulations must account for a modified level scheme and for the realistic trapping potentials and polarization geometries of the physical setup.

Bibliography

[1] Leonardo Razzai. “Setup of a Magneto-Optical Trap of cold 87Rb atoms in proximity to an hollow core photonic crystal fiber”. Relatore: Prof. Francesco Minardi; Correlatore: Prof. Marco Prevedelli. MA thesis. Alma Mater Studiorum – Università di Bologna, 2024.

[2] M. Marchesini. “A new magneto-optical trap for cold rubidium atoms in hollow-core fibres”. PhD thesis. University of Bologna, 2025.

[3] R. Wang *et al.* “Optical trapping of mesoscale particles and atoms in hollow-core optical fibers: principle and applications”. In: *Light: Science & Applications* 14 (2025), p. 146. DOI: 10.1038/s41377-025-01801-5.

[4] Site Editor. *What Do You Know About Polarization Maintaining Optical Fiber?* Newsun Tech, 2022. URL: https://www.ns-fiber.com/new/What-Do-You-Know-About-Polarization-Maintaining-Optical-Fiber.html.

[5] W. J. Riley. *Handbook of Frequency Stability Analysis*. NIST Special Publication 1065. National Institute of Standards and Technology, 2008. DOI: 10.6028/NIST.SP.1065.

[6] B. E. A. Saleh and M. C. Teich. *Fundamentals of Photonics*. 3rd. John Wiley & Sons, 2019. ISBN: 978-1-119-50687-4.

[7] J. R. Johansson, P. D. Nation, and F. Nori. “QuTiP: An open-source Python framework for the dynamics of open quantum systems”. In: *Computer Physics Communications* 183.8 (2012), pp. 1760–1772. DOI: 10.1016/j.cpc.2012.02.021.

[8] Daniel A. Steck. *Rubidium 87 D Line Data*. Available online at http://steck.us/alkalidata. Revision 2.2.3 (9 July 2023). 2001.